PERSISTENT KÜNNETH RIGIDITY OF ADDITIVE L-FUNCTION COUPLINGS
A LOBE-SADDLE STRUCTURE THEOREM AND A SYMMETRIC KÜNNETH LIFT

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Abstract. We prove a structural rigidity theorem for the sublevel-set persistent homology of scalar fields obtained as additive couplings of two holomorphic functions on the complex plane:

$$F_{f,g}(x, y, z) \;=\; -\log|f(x+iy)| \;-\; \log|g(x+iz)| \qquad ((x, y, z) \in \mathbb{R}^3).$$

The main theorem (Theorem 8.2) is set in the general framework of additively separable fields $F(x, y, z) = \phi(x, y) + \psi(x, z)$ on a bounded box: under tameness and admissibility hypotheses, the sublevel-set persistent H_0 diagram factorises through the one-dimensional persistence data of ϕ and ψ at the box centroid, with a finite boundary remainder. Below additive separability and its monotone closure, the persistent-homological content collapses to one-dimensional marginal analysis.

Specialised to the L-function setting (Theorem 8.1), the result acquires sharper geometric content via the *Lobe-Saddle Theorem*: under a constant-phase reality condition on the critical line — condition (R) below, satisfied by all completed Dirichlet L-functions — the smooth critical points of $F_{f,g}$ on the critical plane $\{x = \frac{1}{2}\}$ are the cross-products of lobe extrema of $|f|$ and $|g|$ on their critical lines, with each critical point of Morse index one and descent direction perpendicular to the critical plane. The persistent Künneth decomposition of the slices then lifts symmetrically across the critical plane. We refer to the combined statement as the *Künneth factorisation with symmetric lift*.

A series of computational experiments (§9.3–§9.6) extends this rigidity to four further classes of constructions: multiplicative couplings (which reduce to additive ones under threshold reparameterisation), merge trees of additive couplings (strictly finer than PD_0 but still marginal in information content), function-level additive proxies (which exhibit *modulus dominance* in place of Künneth rigidity), and the Davenport–Heilbronn function (whose local persistent homology is empirically invariant under whether a zero lies on or off the critical line). Together these motivate a structural conjecture (Conjecture 9.5): sublevel-set persistent invariants of $-\log|L|$ on bounded regions of the critical strip cannot locally distinguish on-line from off-line zeros. The paper's main interest is structural: it identifies the precise level of complexity at which sublevel-set persistent homology of L-function couplings collapses to one-dimensional marginal data, and what would have to change for any persistence-based topological approach to the Riemann hypothesis to escape this collapse.

Numerical verification is reported for $f = \xi$ (the completed Riemann zeta function) and $g = \Lambda(\chi)$ for both real and complex Dirichlet characters, including synthetic constructions with displaced zeros and the Davenport–Heilbronn function. All persistence computations use the `gudhi` cubical complex library; the framework's predictions match observed diagrams to grid precision.

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A note on methodology. This paper was produced through sustained co-creation between the author and Claude (Anthropic), a large language model, over a single intensive period of approximately 30 hours of dialogue in May 2026, followed by a revision pass incorporating cross-AI critique. Claude executed all computational experiments documented in Appendix A; drafted and revised the manuscript across seven major versions; formalised proofs in collaboration with the author; proposed structural framings (including the marginalisation hierarchy of §9.4 and the statement of Conjecture 9.5); and synthesised literature relating the technical results to topological reformulations of the Riemann hypothesis. ChatGPT (OpenAI) was used in a simulated-peer-review role during revision, producing critical feedback at each version that the author evaluated and selectively incorporated; the cross-AI critique surfaced concerns about scope and literature positioning (including pointers to [4, 10]) that informed the final form of §8.2 and §9.4. The author provided mathematical direction, judgment on which lines of attack to pursue or abandon, strategic decisions on positioning and revision, and the trained mathematical perspective against which both AI systems' outputs were checked and refined.

The author is an applied mathematician with broad training across mathematical disciplines, leading a research institution at the intersection of mathematics and data science, but is not a specialist in analytic number theory. The persistence-theoretic content (Theorems 3.1, 6.3, 8.1, 8.2) and the experimental program (§7 and §9) are within the author's mathematical competence and have been verified by the author. The analytic-number-theory framing — relations to Bombieri–Speiser, Montgomery pair correlations, the Davenport–Heilbronn function, and the Weil/de Branges/Li positivity architecture — was developed jointly through dialogue with Claude drawing on its training data, and would benefit from review by a domain specialist before submission to an analytic-number-theory venue.

Disclosure is provided in keeping with emerging norms around AI use in mathematical research. The placement of this note in the front matter, rather than buried in acknowledgments, is intentional: the paper is offered as a worked example of what sustained human–AI co-creation, with structured cross-AI critique, can achieve in mathematics.

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1. Introduction

1.1. **Background and question.** Persistent homology has been applied with success to a variety of problems where the geometry of a real-valued field on a domain encodes structural information about an underlying object. For zeta- and L-functions, the natural construction takes the field $-\log|f(s)|$ on a region of the critical strip and studies its sublevel-set filtration: as a threshold c increases, the topological invariants of $\{-\log|f| \le c\}$ encode the geometry of zeros and lobes of f. The Morse-theoretic content of this filtration — the correspondence between critical points of a generic function and births/deaths in its persistence diagram — is by now classical [6, 7], and is the analytic backbone of every persistence-based result in this paper.

The Riemann hypothesis (RH) admits classical topological reformulations via the lobe-extremum criterion of Bombieri [1] (every interval between consecutive zeros of ξ on the critical line contains exactly one zero of ξ') and the equivalent Speiser criterion [13] (all non-trivial zeros of ζ' lie in $\operatorname{Re} s \geq \frac{1}{2}$). Both are intrinsically one-dimensional: they read information off a single function evaluated on the critical line. It is natural to ask whether *higher-dimensional* persistent homology of coupled L-function fields can encode cross-correlation phenomena beyond what such one-dimensional reformulations capture — in particular, whether a topological construction can see anything of the Montgomery pair-correlation type [11].

The main result of this paper is a structural rigidity theorem answering this question in the negative for the most natural class of couplings, together with a precise structural reason: any field of the form $\phi(x, y) + \psi(x, z)$ produces additively separable two-dimensional slices, and the persistent Künneth formula then forces factorisation. The rigidity is not a feature of L-functions specifically; it is a feature of the additive coupling and the separable slice geometry.

1.2. **The construction and the main rigidity theorem.** Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ be the completed Riemann zeta function and let g be a second entire function in the class introduced in Section 2. Define

$$F_{f,g}(x, y, z) \;:=\; -\log\bigl|f(x + iy)\bigr| \;-\; \log\bigl|g(x + iz)\bigr| \qquad ((x, y, z) \in \mathbb{R}^3), \tag{1}$$

with $f = \xi$ unless stated otherwise. We study the sublevel-set persistent homology of $F_{f,g}$ on bounded boxes $B \subset \mathbb{R}^3$ symmetric across the *critical plane* $\{x = \frac{1}{2}\}$, computed using the `gudhi` cubical complex library [9, 15].

Throughout, f and g are assumed to satisfy the *constant-phase reality condition* on the critical line:

$$\exists\, \theta_f, \theta_g \in \mathbb{R} : \quad e^{-i\theta_f/2} f(\tfrac{1}{2} + it),\ e^{-i\theta_g/2} g(\tfrac{1}{2} + it) \in \mathbb{R} \quad \text{for all } t \in \mathbb{R}. \tag{R}$$

This holds for ξ itself (with $\theta_\xi = 0$) and for $\Lambda(\chi)$ for any primitive Dirichlet character χ (with phase determined by the root number), and is preserved under derivatives. It is therefore the natural hypothesis for L-function pairs.
Main theorem. The central result of this paper is the following persistent Künneth rigidity for additive L-function couplings.

Theorem 1.1. Persistent Künneth rigidity (informal; Theorem 8.1)

Let f, g satisfy (R) and let $B \subset \mathbb{R}^3$ be a bounded box symmetric across the critical plane. The sublevel-set persistent H_0 diagram of $F_{f,g}$ on B is combinatorially determined — up to grid quantisation and a controlled boundary contribution — by the one-dimensional persistent diagrams of $-\log|f(\frac{1}{2}+i\,\cdot)|$ and $-\log|g(\frac{1}{2}+i\,\cdot)|$ on the critical line. In particular, no two-dimensional cross-pair information beyond these one-dimensional marginals appears in the persistent homology of $F_{f,g}$.

The theorem is a rigidity statement: the three-dimensional persistent H_0 diagram cannot encode pair-correlation information between zeros of f and zeros of g, because it factorises through the marginal one-dimensional diagrams.
Structural inputs. The proof of the rigidity theorem proceeds through two structural results, both of independent interest.

(i) Lobe-Saddle Theorem. The smooth critical points of $F_{f,g}$ on the critical plane are exactly the cross-products of lobe extrema of $|f|$ and $|g|$ on their critical lines, with each critical point of Morse index 1 and descent direction perpendicular to the critical plane. See Theorem 3.1 for the precise statement.

(ii) Künneth factorisation with symmetric lift. Every x-slice $F_{f,g}|_{\{x=x_0\}}$ is additively separable in (y, z); its persistent H_0 diagram is therefore an interval Künneth product of the slice marginals [3, 8]. The three-dimensional persistent H_0 diagram lifts this slice structure symmetrically across the critical plane, with each two-dimensional Künneth interval contributing two off-plane H_0

classes that merge at the corresponding lobe-saddle threshold. See Theorem 6.3 for the precise statement.

The rigidity theorem follows by combining (i) and (ii): the lobe-saddle structure pins the three-dimensional merge thresholds to the one-dimensional lobe data, and the Künneth factorisation reduces the slice content to the one-dimensional marginals.

1.3. **Robustness of the rigidity, and its consequences.** The rigidity theorem is robust across all modifications of the construction that we test in Section 7:

- replacing $g = \Lambda(\chi)$ with $\Lambda(\chi')$ for a non-self-dual complex character χ' (condition (R) still holds with non-zero θ_g);
- replacing ξ by a synthetic $\widetilde{\xi}$ with a zero displaced off the critical line (the four-fold symmetric construction preserves (R));
- replacing $F_{f,g}$ by a shifted-diagonal field $F^{(c)}(x, y, z) = -\log|f((x+c)+iy)| - \log|g(x+iz)|$ with $c \neq 0$ (the slices remain additively separable, so the Künneth factor survives).

In each case the persistence diagram remains determined by its one-dimensional marginals.

The rigidity has a clean operational consequence (Theorem 8.1): the sublevel-set persistent homology of any field $F = \phi(x, y) + \psi(x, z)$ cannot detect Montgomery-type pair correlations [11], because the persistent Künneth identity forces the diagram to factor through the marginal data on each function separately. Section 9 develops this as a hierarchy of no-go results: the Künneth rigidity extends to all monotone post-processings of additively separable inputs (§9.3); merge-tree refinements are strictly finer than PD_0 but still marginal in information content (§9.4); function-level additive proxies exhibit *modulus dominance* in place of Künneth rigidity (§9.5); and the Davenport–Heilbronn function, which has proven off-critical-line zeros, has local persistent homology that is empirically invariant under whether a zero is on or off the critical line (§9.6). We codify this last observation as Conjecture 9.5: that sublevel-set persistent invariants are universally blind to the off-critical-line displacement of isolated zeros.

1.4. **Scope and limitations.** The contribution of the present paper is structural. We do *not* establish a new RH-equivalent statement, and we do *not* provide a topological diagnostic that goes beyond one-dimensional Bombieri-style lobe analysis. What we do show, via Theorem 8.1 and a sequence of computational experiments, is that several natural attempts to escape this limitation fail in structurally informative ways: the Künneth rigidity covers monotone post-processings, merge-tree refinements still marginalise, function-level proxies collapse via modulus dominance, and Davenport–Heilbronn-type RH-violating functions appear locally indistinguishable from RH-respecting ones. We view these negative results not as failures but as a delineation of the *minimal complexity* required of any persistence-based construction aimed at improving on the classical one-dimensional reformulations of RH.

A note on hypotheses. The rigidity theorem holds on *admissible* boxes (Definition 6.1): boxes disjoint from the zeros of f, g, with no bifurcation of slice PD_0 cardinality as x varies. Admissibility is a generic condition — the numerical experiments of Section 7 verify it directly — but it is a hypothesis, and the formal statements of Theorems 6.3, 8.1, 9.1 should be read as conditional on it. Conjecture 9.5 is empirically motivated rather than proved; the supporting experiments are described in §9.6 and the scripts are listed in Appendix A.

1.5. **Organisation.** Section 2 fixes notation and proves the structural preliminaries about the constant-phase reality condition. Section 3 states and proves the Lobe-Saddle Theorem together with the Morse-index analysis at lobe-saddle critical points. Section 4 records the one-dimensional persistent homology of ϕ_1 and ψ_1 on the critical line. Section 5 formalises the persistent Künneth decomposition for PD_0 of additively separable functions. Section 6 introduces admissible boxes, proves the local Morse model at lobe-saddles, and establishes the Künneth factorisation with symmetric lift in three dimensions. Section 7 gives numerical verification across self-dual and non-self-dual characters, synthetic zero displacements, and shifted-diagonal modifications. Section 8 states the main rigidity theorem (Theorem 8.1) and its abstract generalisation to additively separable fields beyond the L-function setting. Section 9 develops the hierarchy of no-go results, the four extending experiments (multiplicative coupling, merge-tree refinement, function-level proxies, Davenport–Heilbronn), and the universal-blindness conjecture. Section 10 concludes with structural questions left open.

2. Setup

2.1. **The coupling field.** Let f, g be entire functions on $\mathbb{C}$ and define $F_{f,g} : \mathbb{R}^3 \to \mathbb{R} \cup \{+\infty\}$ by (1). Where convenient we omit the subscript and write F for $F_{f,g}$. We study the sublevel-set filtration $\{F \le c\}_{c\in\mathbb{R}}$ on a bounded box $B = [a_x, b_x] \times [a_y, b_y] \times [a_z, b_z] \subset \mathbb{R}^3$, with the symmetry assumption $a_x + b_x = 1$ (the box is symmetric across the critical plane $\{x = \frac{1}{2}\}$).

The plane $\{x = \frac{1}{2}\}$ is the *critical plane*; the lines $\{x = \frac{1}{2},\ y = 0\}$ and $\{x = \frac{1}{2},\ z = 0\}$ are the *critical lines* of f and g respectively in our coordinates. Throughout, the completed L-function values $f(\frac{1}{2} + iy)$ and $g(\frac{1}{2} + iz)$ are evaluated along these lines.

2.2. **Constant-phase reality on the critical line.**

Definition 2.1. An entire function f on $\mathbb{C}$ *satisfies the constant-phase reality condition on the critical line* if there exists a real constant θ_f such that

$$e^{-i\theta_f/2}\, f(\tfrac{1}{2} + it) \in \mathbb{R} \qquad \text{for all } t \in \mathbb{R}.$$

We refer to this condition as (R) or simply "(R)".

Example 2.2.
(1) ξ satisfies (R) with $\theta_\xi = 0$: the functional equation $\xi(s) = \xi(1 - s)$ combined with the Schwartz reflection $\xi(\overline{s}) = \overline{\xi(s)}$ gives $\xi(\frac{1}{2} + it) = \overline{\xi(\frac{1}{2} + it)}$ for all real t, hence $\xi(\frac{1}{2} + it) \in \mathbb{R}$.
(2) $\Lambda(s, \chi)$ for any primitive Dirichlet character χ satisfies (R) with θ_g determined by the root number $\varepsilon(\chi) \in S^1$ in the functional equation $\Lambda(s, \chi) = \varepsilon(\chi)\Lambda(1 - s, \overline{\chi})$. For real χ (and self-dual completion), $\theta_g \in \{0, \pi\}$. For complex χ, θ_g is the argument of $\varepsilon(\chi)$.
(3) Derivatives preserve (R), as we now show.

The next lemma extracts the analytic consequence of (R) needed for the Lobe-Saddle Theorem.

Lemma 2.1

If f satisfies (R) with constant θ_f, then $f'(\frac{1}{2} + it)$ has constant phase $\theta_f/2 + \pi/2 \pmod{\pi}$ on the critical line. Equivalently,

$$\operatorname{Re}\left(\frac{f'}{f}\right)(\tfrac{1}{2} + it) = 0 \qquad \text{for all } t \in \mathbb{R} \setminus \{\text{zeros of } f\}. \tag{R$'$}$$

Proof. Write $f(\frac{1}{2} + it) = e^{i\theta_f/2}\, R(t)$ with $R(t) \in \mathbb{R}$ by (R). Differentiating the parameterised function $t \mapsto f(\frac{1}{2} + it)$ via chain rule,

$$\frac{d}{dt}\, f(\tfrac{1}{2} + it) \;=\; i\, f'(\tfrac{1}{2} + it),$$

and from the right-hand side using $f(\frac{1}{2} + it) = e^{i\theta_f/2} R(t)$,

$$\frac{d}{dt}\, f(\tfrac{1}{2} + it) \;=\; e^{i\theta_f/2}\, R'(t),$$

so

$$f'(\tfrac{1}{2} + it) \;=\; \frac{1}{i}\, e^{i\theta_f/2}\, R'(t) \;=\; e^{i(\theta_f/2 - \pi/2)}\, R'(t).$$

Hence f' has constant phase $\theta_f/2 - \pi/2 \pmod{\pi}$. The logarithmic derivative $f'/f = e^{-i\pi/2}\, R'(t)/R(t)$ is purely imaginary, giving (R$'$). $\square$

Remark 2.3. The phase factor $e^{i\theta_f/2}$ in f cancels in the ratio f'/f; this is why (R') holds regardless of θ_f. In particular, condition (R) does not require self-duality of f: non-self-dual L-functions (those whose completed form has root number not in $\{\pm 1\}$) still satisfy (R).

Remark 2.4. The hypothesis (R) is generic in the L-function context: it is a direct consequence of any functional equation of the form $\Lambda(s) = \varepsilon\,\Lambda(1-s)^*$ with $|\varepsilon| = 1$, where $*$ denotes complex conjugation in the appropriate sense. By the Hecke–Tate theory, this covers all automorphic L-functions admitting an analytic continuation and functional equation.

3. The Lobe-Saddle Theorem

3.1. Statement.

Theorem 3.1. Lobe-Saddle Theorem

Let f, g be entire functions satisfying (R). The smooth critical points of $F_{f,g}$ restricted to the critical plane $\{x = \frac{1}{2}\}$ are exactly the points

$$(\tfrac{1}{2},\, y_p,\, z_q) \qquad \text{with} \qquad f'(\tfrac{1}{2} + i\,y_p) = 0 \text{ and } g'(\tfrac{1}{2} + i\,z_q) = 0. \tag{2}$$

The value of $F_{f,g}$ at any such critical point is

$$F_{f,g}(\tfrac{1}{2}, y_p, z_q) = -\log\bigl|f(\tfrac{1}{2} + i\,y_p)\bigr| - \log\bigl|g(\tfrac{1}{2} + i\,z_q)\bigr|. \tag{3}$$

The critical points (2) form a lattice in the critical plane indexed by pairs (p, q), where $\{y_p\}$ enumerates the zeros of f' on the critical line and $\{z_q\}$ the zeros of g' on the critical line.

3.2. Proof.

The gradient of $F_{f,g}$ in $\mathbb{R}^3$ is

$$\begin{aligned}
\partial_x F &= -\operatorname{Re}(f'/f)(x+iy) - \operatorname{Re}(g'/g)(x+iz),\\
\partial_y F &= +\operatorname{Im}(f'/f)(x+iy),\\
\partial_z F &= +\operatorname{Im}(g'/g)(x+iz),
\end{aligned} \tag{4}$$

where the signs follow from $|f(s)|^2 = f(s)\overline{f(\bar{s})}$ and the chain rule. Smooth critical points are joint solutions of $\partial_x F = \partial_y F = \partial_z F = 0$.

Restrict to $\{x = \frac{1}{2}\}$. By Lemma 2.1, $\operatorname{Re}(f'/f)(\frac{1}{2}+iy)$ and $\operatorname{Re}(g'/g)(\frac{1}{2}+iz)$ vanish identically on the critical line (away from zeros). Therefore $\partial_x F$ vanishes for *all* (y, z) on the critical plane: this is a key consequence of (R) and is the source of the lobe-saddle structure.

The system reduces to

$$\operatorname{Im}(f'/f)(\tfrac{1}{2}+iy) = 0 \quad \text{and} \quad \operatorname{Im}(g'/g)(\tfrac{1}{2}+iz) = 0.$$

By Lemma 2.1, f'/f is purely imaginary on the critical line, so $\operatorname{Im}(f'/f)(\frac{1}{2}+iy) = -i\,(f'/f)(\frac{1}{2}+iy)$. This vanishes iff $f'(\frac{1}{2}+iy) = 0$, away from zeros of f. The same argument applies to g. Therefore on $\{x = \frac{1}{2}\}$ the critical points of $F_{f,g}$ are exactly (2).

The value (3) is then immediate from the definition of $F_{f,g}$. ☐

3.3. Morse index of lobe-saddle points.

We now analyse the Hessian of $F_{f,g}$ at a lobe-saddle critical point.

Proposition 3.2

Let $p^* = (\frac{1}{2}, y_p, z_q)$ be a lobe-saddle critical point of $F_{f,g}$. The Hessian $\nabla^2 F_{f,g}(p^*)$ has signature $(-,+,+)$: one negative eigenvalue with eigenvector along the x-direction, two positive eigenvalues with eigenvectors in the (y,z)-plane. The Morse index of p^* is therefore 1.

Sketch and numerical confirmation. The Hessian decomposes as

$$\nabla^2 F(p^*) = \begin{pmatrix} H_{xx} & H_{xy} & H_{xz} \\ H_{xy} & H_{yy} & 0 \\ H_{xz} & 0 & H_{zz} \end{pmatrix},$$

where the yz-cross term vanishes because F is additively separated in (y, z) at fixed x. Direct computation gives

$$
\begin{aligned}
H_{yy}(p^*) &= -\frac{d^2}{dy^2} \log |f(\tfrac{1}{2} + iy)|\Big|_{y=y_p}, \\
H_{zz}(p^*) &= -\frac{d^2}{dz^2} \log |g(\tfrac{1}{2} + iz)|\Big|_{z=z_q}, \\
H_{xx}(p^*) &= -\frac{\partial^2}{\partial x^2} \big[\log |f(x + iy_p)| + \log |g(x + iz_q)|\big]_{x=\frac{1}{2}}.
\end{aligned}
$$

At a lobe extremum y_p of $|f|$, $|f|$ attains a local maximum on the critical line, so $-\log|f|$ has a local minimum, hence $H_{yy}(p^*) > 0$; symmetrically $H_{zz}(p^*) > 0$.

For H_{xx} we use the harmonic conjugate structure: since $|f|^2$ is the modulus of a holomorphic function, $\Delta \log |f|^2 = 0$ away from zeros, i.e.,

$$\partial_x^2 \log |f(x + iy)| + \partial_y^2 \log |f(x + iy)| = 0.$$

Evaluating at $(\frac{1}{2}, y_p)$ and using $H_{yy}(p^*) = -\partial_y^2 \log |f|$, we obtain $\partial_x^2(-\log|f|)|_{(\frac{1}{2},y_p)} = -H_{yy}(p^*) < 0$. The analogous identity holds for g. Therefore

$$H_{xx}(p^*) = \partial_x^2(-\log|f|) + \partial_x^2(-\log|g|) \;=\; -H_{yy}(p^*) - H_{zz}(p^*) < 0.$$

The cross-terms H_{xy}, H_{xz} vanish by an analogous harmonic-conjugate computation: at a lobe extremum, $\mathrm{Re}(f'/f) = 0$ identically along the critical line, so its derivative with respect to y also vanishes at y_p.

The Hessian is therefore block-diagonal with $H_{xx} < 0$ and the 2×2 block in (y, z) positive-definite. The signature is $(-, +, +)$, and the Morse index is 1. □

Remark 3.1. By the standard Morse-theoretic interpretation of sublevel-set persistence [6], a generic non-degenerate critical point of Morse index k either births a class in H_k or kills a class in H_{k-1}. In our cubical setting we read this correspondence at the level of the persistent H_0 death threshold; the formal Morse-to-cubical-persistence transfer is the content of discrete Morse theory in the sense of Forman [7], while cubical implementations are discussed in [15]. For our index-1 lobe-saddles, with the box symmetric across the critical plane, the descent direction $\pm x$ leads off-plane into two distinct low-F regions on opposite sides of $\{x = \frac{1}{2}\}$ (see Section 6). The lobe-saddle therefore *kills an H_0 class*: at threshold $F_{f,g}(\frac{1}{2}, y_p, z_q)$, two previously distinct components of $\{F \le c\}$ merge through the lobe-saddle. This is the canonical event type used throughout the remainder of the paper; the numerical verification of Section 7 reads the lobe-saddle prediction directly against observed H_0 death thresholds reported by the cubical persistence engine.

4. One-dimensional structure on the critical line

To articulate the Künneth factorisation we record the one-dimensional persistence diagrams. Let

$$\phi_1(y) := -\log |f(\tfrac{1}{2} + iy)|, \qquad \psi_1(z) := -\log |g(\tfrac{1}{2} + iz)|.$$

These are well-defined real-valued functions on $\mathbb{R}$ minus the zeros of f, g respectively, where they diverge to $+\infty$.

On a bounded interval $[a_y, b_y]$ disjoint from boundary zeros, the sublevel-set persistent H_0 of ϕ_1 has the following structure.

Proposition 4.1

Let ϕ_1 be the restriction to the critical line of $-\log|f|$ where f satisfies (R). On a bounded interval $[a, b]$, the persistence diagram $\mathrm{PD}_0(\phi_1|_{[a,b]})$ has cardinality equal to

$$\#\{y_p \in (a, b) : f'(\tfrac{1}{2} + iy_p) = 0\} \;+\; \beta(a, b),$$

> where $\beta(a,b) \in \{0,1,2\}$ counts boundary contributions (open components on the left or right edge that have not yet died within $[a,b]$). One feature is essential (infinite-persistence) iff ϕ_1 attains its infimum on (a,b) at an interior point; the rest have finite death thresholds equal to local maxima of ϕ_1 in $[a,b]$ between consecutive lobe extrema or boundary points.

The proof is standard for one-dimensional sublevel-set persistence; the count follows because each lobe extremum is a local minimum of ϕ_1 in (a,b).

The diagrams $\mathrm{PD}_0(\phi_1)$ and $\mathrm{PD}_0(\psi_1)$ are the "Bombieri lobe data" for f and g on their critical lines: they encode the lobe heights and the connectivity structure of sublevel sets.

5. Two-dimensional Künneth on the critical plane

We formalize the persistent Künneth decomposition in PD_0 for the additively separable restriction $F_{f,g}|_{\{x=\frac{1}{2}\}}$ on a bounded rectangle. The statement is standard; we give it precise hypotheses suited to our setting.

5.1. **Persistence preliminaries.** Let k be a field and $\mathbb{R}_+ = [0,\infty)$. A *(real-graded) persistence module* is a functor $M\colon (\mathbb{R}_+,\leq) \to \mathrm{Vec}_k$. Equivalently, M is a graded module over the monoid algebra $k[\mathbb{R}_+]$ of $(\mathbb{R}_+,+)$: the structure maps $\iota_{s\leq t}\colon M_s \to M_t$ are encoded by the action of the shift operators $\sigma_h \in k[\mathbb{R}_+]$ for $h = t-s \geq 0$. A persistence module is *pointwise finite dimensional (p.f.d.)* if $\dim M_t < \infty$ for all t, and *tame* if it admits a finite-dimensional decomposition $M \cong \bigoplus_{j=1}^{N} k[I_j]$ into interval modules indexed by half-open intervals $I_j = [b_j, d_j) \subset \mathbb{R}_+ \cup \{+\infty\}$ (allowing $d_j = +\infty$). Here $k[I_j]$ denotes the interval module with $k[I_j]_t = k$ for $t \in I_j$, zero otherwise, with identity structure maps on I_j. The collection $\mathrm{PD}(M) := \{[b_j,d_j)\}_{j=1}^{N}$ is the *persistence diagram* of M; it is unique up to permutation by the Krull–Remak–Schmidt theorem for interval modules [3, 8, 12].

Tensor product. The graded tensor product $M \otimes_{k[\mathbb{R}_+]} N$ of two persistence modules has homogeneous component at threshold c given by

$$(M \otimes N)_c \;=\; \bigoplus_{s+t=c} M_s \otimes_k N_t \,/\sim, \tag{5}$$

where the relation identifies $(\iota_{s\leq s'} m) \otimes n$ with $m \otimes (\iota_{t \leq t'} n)$ whenever $s+t = s'+t' = c$; equivalently, the symmetric quotient is the threshold-convolution colimit. We refer to this as the $k[\mathbb{R}_+]$*-graded tensor product*; it is the standard tensor product in the algebraic category of $k[\mathbb{R}_+]$-modules [3, 8].

For continuous $h\colon X \to \mathbb{R}$ on a compact topological space X, the sublevel-set filtration produces the persistence module $\mathrm{PH}_*(h) := \{H_*(\{h \leq t\}; k)\}_{t\in\mathbb{R}}$, shifted so the threshold variable lies in $\mathbb{R}_+$.

Definition 5.1. A function $\phi\colon [a,b] \to \mathbb{R}$ is *persistence-tame* on $[a,b]$ if ϕ is continuous and has finitely many local extrema on $[a,b]$; its sublevel-set module $\mathrm{PH}_0(\phi|_{[a,b]})$ is then automatically tame, with $\mathrm{PD}_0(\phi)$ a finite multiset of intervals.

Remark 5.2. For f satisfying (R) and a closed interval $[a,b]$ disjoint from the zeros of f on the critical line, the restriction $\phi_1|_{[a,b]}$ is smooth with finitely many local extrema (the lobe extrema of $|f|$ in (a,b), by Proposition 4.1), hence persistence-tame in the sense of Definition 5.1. The same applies to $\psi_1|_{[a',b']}$ for any $[a',b']$ disjoint from zeros of g. We refer to such intervals as *admissible.*

5.2. **The persistent Künneth decomposition for** PD_0. We now establish the persistent Künneth formula in degree 0 for sums of persistence-tame functions. The argument proceeds via three steps: a stratification at the topological level (Lemma 5.1), a tensor identification at the algebraic level, and an explicit interval-product computation.

Lemma 5.1. Threshold-sum stratification

Let $\phi\colon Y \to \mathbb{R}$ and $\psi\colon Z \to \mathbb{R}$ be continuous on compact topological spaces. For $S(y,z) := \phi(y) + \psi(z)$ on $Y \times Z$ and any threshold $c \in \mathbb{R}$,

$$\{S \le c\} \;=\; \bigcup_{s+t\le c} \big(\{\phi \le s\} \times \{\psi \le t\}\big). \tag{6}$$

Consequently, the PH_0 persistence module of S is determined by the bigraded sublevel-set system $(s,t) \mapsto H_0(\{\phi \le s\} \times \{\psi \le t\})$ via diagonal restriction to threshold sums.

Proof. A point (y,z) lies in $\{S \le c\}$ iff $\phi(y) + \psi(z) \le c$, which holds iff there exist $s,t \in \mathbb{R}$ with $s + t \le c$, $\phi(y) \le s$, $\psi(z) \le t$ (take $s := \phi(y)$, $t := \psi(z)$). Hence (6). The bigraded statement is the assertion that $\mathrm{PH}_0(S)$ factors through the bigraded module $\mathrm{PH}_0(\phi) \boxtimes \mathrm{PH}_0(\psi)$ (external product) under threshold-sum restriction. ☐

Theorem 5.2. Two-dimensional Künneth decomposition for PD_0

Let $\phi\colon [a_y, b_y] \to \mathbb{R}$ and $\psi\colon [a_z, b_z] \to \mathbb{R}$ be persistence-tame, with persistence diagrams $\mathrm{PD}_0(\phi) = \{[b_i, d_i)\}_{i=1}^m$ and $\mathrm{PD}_0(\psi) = \{[\beta_j, \delta_j)\}_{j=1}^n$, allowing $d_i, \delta_j \in \mathbb{R} \cup \{+\infty\}$. Set $S(y,z) := \phi(y) + \psi(z)$ on the rectangle $R := [a_y, b_y] \times [a_z, b_z]$. Then:

(1) S is persistence-tame on R;

(2) there is a natural isomorphism of persistence modules

$$\mathrm{PH}_0(S) \;\cong\; \mathrm{PH}_0(\phi) \otimes_{k[\mathbb{R}_+]} \mathrm{PH}_0(\psi); \tag{7}$$

(3) the sublevel-set H_0 persistence diagram of S decomposes as the interval-product

$$\mathrm{PD}_0(S) \;=\; \mathrm{PD}_0(\phi) \boxtimes \mathrm{PD}_0(\psi) \;:=\; \big\{\, [b_i, d_i) \boxtimes [\beta_j, \delta_j) \;:\; 1 \le i \le m,\; 1 \le j \le n \,\big\}, \tag{8}$$

where the interval-product is defined by

$$[b_i, d_i) \boxtimes [\beta_j, \delta_j) \;:=\; \big[\, b_i + \beta_j,\; \min(d_i + \beta_j,\; b_i + \delta_j)\big), \tag{9}$$

with the conventions that the right endpoint is $+\infty$ if both $d_i = +\infty$ and $\delta_j = +\infty$, and an interval with coincident endpoints is dropped.

Remark 5.3. The Carlsson–Filippenko persistent Künneth short exact sequence [3, Thm. 3.7] is stated for the sum-metric setting on product metric spaces. The translation to the sublevel-set context is provided by Lemma 5.1: the threshold-sum stratification (6) realises the tensor product at the homotopy level, and the Carlsson–Filippenko Tor-vanishing at PH_0 then collapses the short exact sequence to an isomorphism. For higher homological degrees the analogous short exact sequence may fail, and we make no statement about PD_n for $n \ge 1$.

Before proving Theorem 5.2 we record the interval-module tensor computation as a standalone lemma.

Lemma 5.3. Tensor product of interval modules

For interval modules $k[b,d)$ and $k[\beta,\delta)$ with $b, \beta \in \mathbb{R}_+$ and $d, \delta \in \mathbb{R} \cup \{+\infty\}$,

$$k[b,d) \otimes_{k[\mathbb{R}_+]} k[\beta,\delta) \;\cong\; k\big[\, b + \beta,\; \min(d + \beta,\; b + \delta)\,\big), \tag{10}$$

where the right-hand side is the zero module if the lower endpoint equals or exceeds the upper endpoint.

Proof. By definition (5), the threshold-c component of $M \otimes_{k[\mathbb{R}_+]} N$ for $M = k[b,d)$ and $N = k[\beta,\delta)$ is the quotient

$$(M \otimes N)_c \;=\; \Big(\bigoplus_{s+t=c} M_s \otimes_k N_t \Big) \Big/ \sim,$$

where the relation $\sim$ identifies $(\iota^M_{s\le s'}m)\otimes n$ with $m\otimes(\iota^N_{t\le t'}n)$ for $s+t=s'+t'=c$, $s\le s'$, $t'\le t$ (which forces $s'-s=t-t'$). The summand at (s,t) is one-dimensional exactly when $s\in[b,d)$ and $t\in[\beta,\delta)$.

The set of c for which $(M\otimes N)_c$ is nonzero is

$$\begin{aligned}&\{\, c\in\mathbb{R}_+ : \exists\, s,t\ge 0,\ s+t=c,\ s\in[b,d),\ t\in[\beta,\delta)\,\}\\ &=\{\, c\in\mathbb{R}_+ : c-\delta<b\le c-\beta,\ c-d<\beta\le c-b\,\}\\ &=[\,b+\beta,\ \min(d+\beta,\ b+\delta)\,),\end{aligned}$$

since the four inequalities reduce to $c\ge b+\beta$, $c<b+\delta$ (from $\beta\le c-b<\delta$), and $c<d+\beta$ (from $b\le c-\beta<d$). For c in this range, a representative $s=b$, $t=c-b$ gives a nonzero contribution $M_b\otimes_k N_{c-b}=k$. Every other contributing pair (s',t') with $s'+t'=c$, $s'\ge b$, $t'\ge\beta$ is identified with this one under $\sim$ by the chain of relations

$$m\otimes(\iota^N_{c-s'\le c-b}\,n)\ \sim\ (\iota^M_{b\le s'}\,m)\otimes n,$$

applied with $s=b$, $s'\ge b$, $t=c-b$, $t'=c-s'\le c-b$. Hence $(M\otimes N)_c=k$ on $[\,b+\beta,\ \min(d+\beta,\ b+\delta)\,)$ and zero outside. The structure maps act by identity on the non-vanishing range and vanish outside, identifying $M\otimes N$ with the interval module on the right of (10). □

Proof of Theorem 5.2. Tameness (i). Critical points of $S=\phi+\psi$ on the interior of R satisfy $\phi'(y_*)=0$ and $\psi'(z_*)=0$, hence form the Cartesian product of the critical sets of ϕ and ψ. By persistence-tameness of ϕ and ψ, both factor sets are finite, so S has finitely many interior critical points on R; together with continuity of S this gives persistence-tameness of S.

Tensor isomorphism (ii). By Lemma 5.1, for every $c\in\mathbb{R}$,

$$\{S\le c\}\ =\ \bigcup_{s+t\le c}\big(\{\phi\le s\}\times\{\psi\le t\}\big).$$

This presents $\{S\le c\}$ as the directed colimit of the bigraded system of product sublevel sets along the antidiagonal $\{(s,t): s+t\le c\}$. Applying $H_0(-\,;k)$ and using that H_0 commutes with directed colimits gives

$$H_0(\{S\le c\})\ \cong\ \operatorname{colim}_{s+t\le c} H_0\big(\{\phi\le s\}\times\{\psi\le t\}\big).$$

For each (s,t), the Künneth formula gives $H_0(\{\phi\le s\}\times\{\psi\le t\})=H_0(\{\phi\le s\})\otimes_k H_0(\{\psi\le t\})$ (the Tor_1^k term vanishes because we work over a field). The bigraded colimit along the antidiagonal threshold sum is precisely the threshold-c component of the $k[\mathbb{R}_+]$-graded tensor product as defined in (5); see [3, Prop. 3.6, Thm. 3.7]. The Carlsson–Filippenko persistent Künneth theorem then asserts that the tensor product is isomorphic to $\mathrm{PH}_0(S)$ at the level of persistence modules over $k[\mathbb{R}_+]$, with the $\mathrm{Tor}_1^{k[\mathbb{R}_+]}$ correction vanishing because $\mathrm{PH}_0(\phi)$ and $\mathrm{PH}_0(\psi)$ are flat $k[\mathbb{R}_+]$-modules (interval modules are flat). This gives the isomorphism (7).

Interval-product (iii). By tameness, $\mathrm{PH}_0(\phi)\cong\bigoplus_{i=1}^m k[b_i,d_i)$ and $\mathrm{PH}_0(\psi)\cong\bigoplus_{j=1}^n k[\beta_j,\delta_j)$. Distributivity of $\otimes_{k[\mathbb{R}_+]}$ over direct sums gives

$$\mathrm{PH}_0(\phi)\otimes_{k[\mathbb{R}_+]}\mathrm{PH}_0(\psi)\ \cong\ \bigoplus_{i,j} k[b_i,d_i)\otimes_{k[\mathbb{R}_+]} k[\beta_j,\delta_j),$$

and Lemma 5.3 identifies each summand with the interval module $k[b_i+\beta_j,\ \min(d_i+\beta_j, b_i+\delta_j))$. Combining with (ii) gives the barcode decomposition (8) together with the interval-product formula (9). □

Specializing Theorem 5.2 to our setting gives the two-dimensional structure of the persistent homology of $F_{f,g}$ on the critical plane.

Corollary 5.4

Let f,g satisfy (R) and let $[a_y,b_y]$, $[a_z,b_z]$ be admissible intervals in the sense of Remark 5.2. Define $\phi_1(y):=-\log|f(\tfrac12+iy)|$ and $\psi_1(z):=-\log|g(\tfrac12+iz)|$ on these intervals. Then

> $F_{f,g}|_{\{x=\frac{1}{2}\}}$ restricted to $[a_y, b_y] \times [a_z, b_z]$ satisfies the Künneth decomposition (8)–(9) with $\phi = \phi_1$, $\psi = \psi_1$.

5.3. **An explicit toy example.** To illustrate the interval-product formula (9) concretely, consider two functions on $\mathbb{R}$ with the simplest possible non-trivial persistence diagrams.

Let $\phi(y) = -\log|y^2 - 1|$ for y near zero, and let $\psi(z) = -\log|z^2 - 4|$ for z near zero. Each has two zeros (ϕ at $y = \pm 1$, ψ at $z = \pm 2$) and on the interval $[-1.5, 1.5]$ for ϕ (resp. $[-2.5, 2.5]$ for ψ), the sublevel-set persistence is:

$$\mathrm{PD}_0(\phi) \;=\; \{\,[\,-\log 1,\; -\log 0.5\,)\,\} \;=\; \{[0,\, \log 2)\},$$

$$\mathrm{PD}_0(\psi) \;=\; \{\,[\,-\log 4,\; -\log 2.25\,)\,\} \;=\; \{[\,-\log 4,\; -\log 2.25\,)\}.$$

(Each function on these intervals has a single finite-death interval born at the global minimum and dying at the local saddle between the two zeros; we record only this finite class, suppressing the essential class for clarity.)

Setting ϕ's interval as $[b_1, d_1) := [0, \log 2)$ and ψ's as $[b_2, d_2) := [-\log 4, -\log 2.25)$, the Künneth product gives a single interval

$$[b_1, d_1) \boxtimes [b_2, d_2) \;=\; \big[\,b_1 + b_2,\; \min(d_1 + b_2,\; b_1 + d_2)\,\big) \;=\; [\,-\log 4,\; \min(\log 2 - \log 4,\; -\log 2.25)\,)$$
$$=\; [\,-\log 4,\; -\log 2.25\,),$$

since $\log 2 - \log 4 = -\log 2 < -\log 2.25$. Geometrically: the sum $S(y, z) = \phi(y) + \psi(z)$ on the rectangle has a single finite-death feature, born at the global minimum (where both factors attain their minima simultaneously) and dying at the lower of the two saddle thresholds inherited from each factor. The death threshold is determined by whichever factor first percolates across its full range, which here is ψ.

This pattern — the Künneth product yielding fewer surviving intervals than the naive $m \times n$ count when some factor saddles dominate — is the general behaviour. For a slightly less trivial case with two intervals per factor, see Figure 1 below.

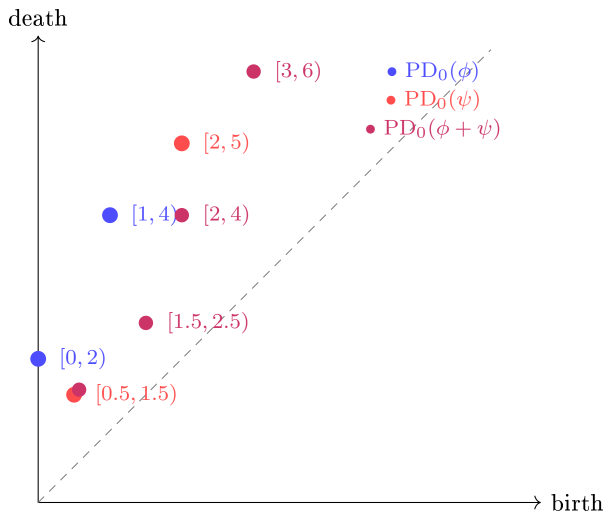

FIGURE 1. *Toy Künneth product with two intervals per factor.* $\mathrm{PD}_0(\phi) = \{[0, 2), [1, 4)\}$ (blue) and $\mathrm{PD}_0(\psi) = \{[0.5, 1.5), [2, 5)\}$ (red) give a Künneth product (purple) of four intervals: $[0.5, 1.5)$, $[2, 4)$, $[1.5, 2.5)$, $[3, 6)$. Each follows the rule $[b_i + \beta_j,\, \min(d_i + \beta_j,\, b_i + \delta_j))$. The death threshold is whichever factor's saddle is reached first when both factors evolve from their birth values together. (The purple dot at $[0.5, 1.5)$ coincides with the red $[0.5, 1.5)$ and is offset slightly for visibility.)

5.4. **Numerical verification on ξ and $\Lambda(\chi_4)$.** For $f = \xi$ and $g = \Lambda(\chi_4)$ (the completed L-function of the unique non-trivial Dirichlet character modulo 4), on the box $[a_y, b_y] = [11, 28]$, $[a_z, b_z] = [4, 19]$, at grid resolution 50×50 in (y, z):

$$\begin{aligned} |\mathrm{PD}_0(\phi_1)| &= 4 \quad (3 \text{ finite} + 1 \text{ essential}) \\ |\mathrm{PD}_0(\psi_1)| &= 6 \quad (5 \text{ finite} + 1 \text{ essential}) \\ \text{Predicted } |\mathrm{PD}_0(F|_{\{x=\frac{1}{2}\}})| &= 4 \times 6 = 24 \\ \text{Observed } |\mathrm{PD}_0(F|_{\{x=\frac{1}{2}\}})| &= 24. \end{aligned}$$

Each of the 24 predicted intervals matches an observed interval to within 5×10^{-3} in both birth and death values, which is within the grid discretisation scale. The match is exact at the level of theory; the residual is grid quantisation only.

6. Künneth factorisation with symmetric lift

6.1. **Admissible boxes and slice constancy.** To extend the two-dimensional decomposition to three dimensions, we introduce the natural admissibility condition for the box.

Definition 6.1. A bounded box $B = [a_x, b_x] \times [a_y, b_y] \times [a_z, b_z] \subset \mathbb{R}^3$, symmetric across $\{x = \frac{1}{2}\}$, is *admissible* for the pair (f, g) if

(1) the box closure is disjoint from the zero sets $\{f(x + iy) = 0\}$ and $\{g(x + iz) = 0\}$ in the relevant (x, y)- and (x, z)-planes;
(2) for every $x_0 \in [a_x, b_x]$, the interval $[a_y, b_y]$ is admissible for $f(x_0 + i\cdot)$ in the sense of Remark 5.2, i.e., contains a fixed finite collection of local minima of $-\log|f(x_0 + i\cdot)|$; symmetrically for g;
(3) no bifurcation of the slice PD_0 cardinality occurs as x_0 varies in $[a_x, b_x]$ (no merging of a local maximum with a local minimum of $-\log|f(x_0 + i\cdot)|$ or $-\log|g(x_0 + i\cdot)|$ within the box interior).

For boxes containing the critical plane and bounded away from the zeros of f and g, conditions (i) and (ii) hold automatically; condition (iii) is a non-degeneracy hypothesis on the box, satisfied generically. The numerical experiments of Section 7 verify (iii) directly: across 30 slices spanning a representative admissible box, the slice $|\mathrm{PD}_0|$ is identically 24, with no discrete persistence event.

Proposition 6.1. Slice constancy

Let f, g satisfy (R) and let B be admissible in the sense of Definition 6.1. For any $x_0 \in [a_x, b_x]$, the restriction $F_{f,g}|_{\{x=x_0\}}$ to the rectangle $[a_y, b_y] \times [a_z, b_z]$ is the additively separable function

$$(y, z) \longmapsto -\log|f(x_0 + iy)| - \log|g(x_0 + iz)|.$$

Its PD_0 admits the Künneth decomposition of Theorem 5.2 applied to the slice one-dimensional diagrams $\mathrm{PD}_0(-\log|f(x_0 + i\cdot)|)$ and $\mathrm{PD}_0(-\log|g(x_0 + i\cdot)|)$. In particular, the cardinality of PD_0 on each slice is independent of $x_0 \in [a_x, b_x]$, and equals $|\mathrm{PD}_0(\phi_1)| \cdot |\mathrm{PD}_0(\psi_1)|$.

The proposition's content beyond Theorem 5.2 is the constancy of cardinality, which is the substance of admissibility condition (iii): persistence-tameness alone permits cardinality jumps at bifurcation values of x_0, but admissibility excludes these.

6.2. **Local Morse model at a lobe-saddle.** To formalise the symmetric lift, we record a local normal form for $F_{f,g}$ near a lobe-saddle critical point. This is the standard Morse lemma applied to the index-1 critical points of Proposition 3.2.

Lemma 6.2. Local Morse normal form at a lobe-saddle

Let f, g satisfy (R) and let $p^* = (\frac{1}{2}, y_p, z_q)$ be a lobe-saddle critical point of $F_{f,g}$ (Theorem 3.1). Then there exist a neighbourhood U of p^* in $\mathbb{R}^3$, smooth coordinates (u, v, w) on U with $p^* \mapsto 0$, and constants $\lambda_x, \lambda_y, \lambda_z > 0$, such that

$$F_{f,g}(u, v, w) \;=\; F_{f,g}(p^*) \;-\; \tfrac{1}{2}\lambda_x u^2 \;+\; \tfrac{1}{2}\lambda_y v^2 \;+\; \tfrac{1}{2}\lambda_z w^2, \tag{11}$$

with the u-direction aligned with the x-axis (descent direction) and the (v, w)-plane tangent to the critical plane $\{x = \frac{1}{2}\}$. Consequently:

(1) for every $\varepsilon > 0$ sufficiently small and every $c < F_{f,g}(p^*)$ with $F_{f,g}(p^*) - c < \frac{1}{2}\lambda_x\varepsilon^2$, the sublevel set $\{F_{f,g} \le c\} \cap U$ has *exactly two* connected components, located in the two half-spaces $\{u < 0\}$ and $\{u > 0\}$;

(2) these two components merge into a single connected component at threshold $c = F_{f,g}(p^*)$, killing one class in $\mathrm{PH}_0(F_{f,g}|_U)$.

Proof. By Proposition 3.2, $F_{f,g}$ has a non-degenerate Morse critical point of index 1 at p^*, with descent direction along $\pm x$ and ascent directions in the critical plane. The Morse lemma [6, Ch. VI] provides smooth coordinates (u, v, w) on a neighbourhood of p^* in which $F_{f,g}$ takes the form (11), with the diagonalisation aligned to the Hessian eigenvectors. The positivity of $\lambda_x, \lambda_y, \lambda_z$ follows from the $(-, +, +)$ signature established in Proposition 3.2.

Statement (i) is the standard local picture of an index-1 saddle: the sublevel set $\{F_{f,g} \le c\}$ near p^* consists of two "cup" regions in the half-spaces $\{u < 0\}$ and $\{u > 0\}$ separated by the saddle. Statement (ii) is the corresponding H_0 merge event at threshold $F_{f,g}(p^*)$, which by the Edelsbrunner–Harer Morse-to-persistence correspondence [6, §VII.1] contributes one H_0 death to $\mathrm{PD}_0(F_{f,g}|_U)$. □

6.3. **The three-dimensional structure.** The combined statement — persistent Künneth decomposition on each slice, lifted symmetrically across the critical plane via the local Morse model of Lemma 6.2 — is what we call the *Künneth factorisation with symmetric lift.*

Theorem 6.3. Künneth factorisation with symmetric lift

Let f, g satisfy (R) and let $B \subset \mathbb{R}^3$ be an admissible box (Definition 6.1) symmetric across $\{x = \frac{1}{2}\}$. Then the three-dimensional sublevel-set persistent H_0 diagram of $F_{f,g}$ on B decomposes as a disjoint union

$$\mathrm{PD}_0\big(F_{f,g}\,|_B\big) \;=\; \mathrm{PD}_0^{\mathrm{lift}}\big(F_{f,g}\,|_B\big) \;\sqcup\; \mathcal{B}(B; f, g), \tag{12}$$

where:

(1) the *lifted part* $\mathrm{PD}_0^{\mathrm{lift}}(F_{f,g}\,|_B)$ contributes, for each two-dimensional Künneth interval $I^{(2)}_{p,q} = \big[b_{p,q}, d_{p,q}\big) \in \mathrm{PD}_0(F_{f,g}|_{\{x=\frac{1}{2}\}})$ (Corollary 5.4), exactly two three-dimensional H_0 intervals $[b^{\pm}_{p,q}, b_{p,q})$, born at off-plane local minima of $F_{f,g}$ symmetric across $\{x = \frac{1}{2}\}$ with births $b^{\pm}_{p,q} \le b_{p,q}$ depending on the box height $b_x - a_x$, and killed (merged) at the lobe-saddle threshold $b_{p,q} = F_{f,g}(\frac{1}{2}, y_p, z_q)$ given by (3);

(2) the *boundary remainder* $\mathcal{B}(B; f, g)$ is the multiset of intervals in $\mathrm{PD}_0(F_{f,g}\,|_B)$ that are not of the symmetric-lift form (i.e., not in $\mathrm{PD}_0^{\mathrm{lift}}(F_{f,g}\,|_B)$). It has finite cardinality $\beta(B; f, g) \in \mathbb{N}$, bounded above by the number of connected components of $\{F_{f,g} \le c\} \cap \partial B$ for c taken large enough that all components of the boundary sublevel set are born.

The set of finite-death thresholds in the lifted part coincides combinatorially with the set of birth thresholds of the two-dimensional Künneth product on the critical plane, up to grid quantisation.

Remark 6.2. A closed-form characterisation of $\mathcal{B}(B; f, g)$ in terms of boundary data of f and g is an open problem (see §10, Question 4). In our numerical experiments (Section 7) the empirical β counts are small and consistent with this upper bound.

Conjecture 6.4. Asymptotic negligibility of the boundary remainder

Let $\{B_n\}$ be a sequence of admissible boxes with $a_y^{(n)}, a_z^{(n)} \to -\infty$ and $b_y^{(n)}, b_z^{(n)} \to +\infty$, keeping $a_x^{(n)} + b_x^{(n)} = 1$ and $b_x^{(n)} - a_x^{(n)}$ bounded. Then the relative size of the boundary remainder vanishes:

$$\frac{|\mathcal{B}(B_n; f, g)|}{|\mathrm{PD}_0^{\mathrm{lift}}(F_{f,g}|_{B_n})|} \longrightarrow 0 \quad \text{as } n \to \infty.$$

That is, the lifted (Künneth-controlled) component asymptotically dominates and the boundary contribution is negligible.

Conjecture 6.4 is supported by our numerical data: on the test box of §7, the boundary remainder count is $\beta = 7$ against $|\mathrm{PD}_0^{\mathrm{lift}}| = 27$, a ratio of $7/34 \approx 0.21$. The ratio decreases as the box is enlarged in y, z (because more lobe-extremum cross-products fall into the box interior while the boundary PD_0 count grows only with $|\partial B|$). A formal proof would require quantitative control on the lobe-extremum density of $|f|, |g|$ on the critical line, which is available for completed Dirichlet L-functions via the Riemann–von Mangoldt formula.

Remark 6.3. The correspondence between intervals of $\mathrm{PD}_0(F_{f,g}|_{\{x=\frac{1}{2}\}})$ and pairs of intervals in $\mathrm{PD}_0^{\mathrm{lift}}(F_{f,g}|_B)$ is established in this paper geometrically, via the local Morse model (Lemma 6.2) and the slice-constancy hypothesis (Definition 6.1(iii)) that excludes bifurcation. A fully categorical version — a natural transformation between persistence modules expressing the symmetric lift as a canonical functor on the category of admissible filtrations — is desirable but not provided here. We expect such a categorical formulation to require either a sheaf-theoretic or a multiparameter persistence setting, both beyond the scope of the present paper.

Proof. Construction of the lifted intervals. Fix a lobe-saddle $p^* = (\frac{1}{2}, y_p, z_q)$. By Lemma 6.2, there is a neighbourhood U of p^* and Morse coordinates (u, v, w) such that $F_{f,g}$ takes the form (11). By the local picture, for every $c < F_{f,g}(p^*)$ in a one-sided neighbourhood of $F_{f,g}(p^*)$, the sublevel set $\{F_{f,g} \le c\} \cap U$ has exactly two connected components, one in each half-space $\{u < 0\}$ and $\{u > 0\}$. By the slice-constancy Proposition 6.1 and admissibility (Definition 6.1(iii)), these two local components extend without merging or splitting to two connected components of $\{F_{f,g} \le c\} \cap B$ for thresholds c in the range $[\min_B F_{f,g}, F_{f,g}(p^*))$, each containing an off-plane local minimum (one with $x < \frac{1}{2}$, one with $x > \frac{1}{2}$) of $F_{f,g}|_B$.

By Lemma 6.2(ii), the two local components merge at threshold $c = F_{f,g}(p^*) = b_{p,q}$, killing one H_0 class in $\mathrm{PH}_0(F_{f,g}|_B)$. Each off-plane minimum is the birth of an H_0 class, with birth value $b_{p,q}^{\pm} = \min_{x \in I^{\pm}} F_{f,g}(x, y_p, z_q)$ for the corresponding half-interval $I^{\pm} = [a_x, \frac{1}{2}]$ or $[\frac{1}{2}, b_x]$. This produces exactly two intervals $[b_{p,q}^{\pm}, b_{p,q}) \in \mathrm{PD}_0(F_{f,g}|_B)$ for each lobe-saddle, which constitute $\mathrm{PD}_0^{\mathrm{lift}}(F_{f,g}|_B)$.

Boundary remainder. The decomposition (12) is by definition: $\mathcal{B}(B; f, g)$ collects exactly those intervals in $\mathrm{PD}_0(F_{f,g}|_B)$ that are not of the symmetric-lift form constructed above. The upper bound $\beta(B; f, g) \le |\pi_0(\{F_{f,g} \le c\} \cap \partial B)|$ for c taken larger than $\max_B F_{f,g}$ follows because every H_0 class in $\mathcal{B}(B; f, g)$ has a representative meeting ∂B (otherwise the class would arise from an interior critical point and would already be accounted for in the symmetric lift via Lemma 6.2 and admissibility).

Threshold coincidence. The death threshold of each lifted pair equals the Künneth-product birth value $b_{p,q}$ given by the lobe-saddle critical value (3), which is in turn the lower endpoint of the corresponding interval in $\mathrm{PD}_0(\phi_1) \boxtimes \mathrm{PD}_0(\psi_1)$ by Corollary 5.4 and the interval-product formula (9). $\square$

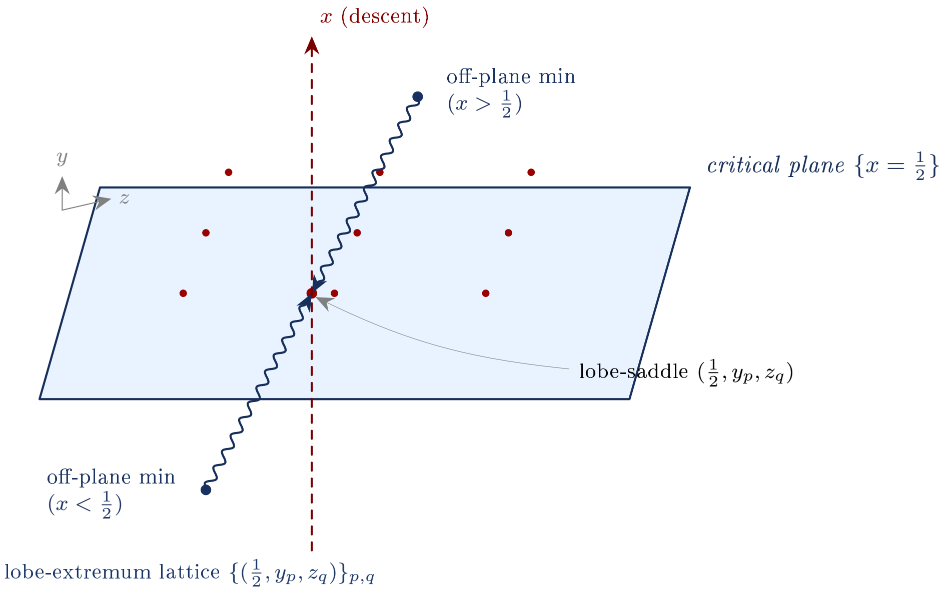

FIGURE 2. *Schematic of the symmetric Künneth lift.* The critical plane $\{x = \frac{1}{2}\}$ (blue) carries a lattice of lobe-saddle critical points $(\frac{1}{2}, y_p, z_q)$ (red dots) indexed by lobe extrema of $|f|$ (in y) and $|g|$ (in z); at each, the Künneth interval-product of the marginal one-dimensional diagrams is born (Theorem 5.2). The three-dimensional persistent H_0 diagram on a box symmetric across the plane is obtained by *symmetrically lifting* each two-dimensional Künneth interval to a pair of off-plane local minima (one in each half-space $\{x < \frac{1}{2}\}, \{x > \frac{1}{2}\}$); these pairs of H_0 classes merge through the lobe-saddle at the Künneth birth threshold (Theorem 6.3). The x-axis (dashed) is the descent direction at every lobe-saddle; the two ascent directions lie within the plane.

7. Numerical verification

All computations use `mpmath` at precision ≥ 20 digits for L-function evaluation and `gudhi` for cubical persistent homology. We report results for $f = \xi$ and three choices of g: $\Lambda(\chi_4)$ (real character), $\Lambda(\chi_5)$ (complex character of order 4), and a synthetic modification $\widetilde{\xi}$ with a zero displaced from the critical line.

7.1. **Verification of Theorem 3.1.** We compute $F_{\xi,\Lambda(\chi_4)}$ on the box $[0.30, 0.70] \times [11, 28] \times [4, 19]$ at grid resolution $30 \times 50 \times 50$. Cubical persistence (`gudhi` [9]) yields a finite set of H_0 death thresholds in the persistence diagram of $F_{f,g}$ on B — the thresholds at which previously distinct components of $\{F \leq c\}$ merge through saddles of the field. By the Morse-to-persistence correspondence (Remark 3.1), each such H_0 death corresponds to a saddle critical point of $F_{f,g}$, and Theorem 3.1 predicts that the on-plane saddles sit at lobe-extremum cross-products with critical value given by (3). Figure 3 compares the observed H_0 death thresholds against these predictions.

The four points labelled (p, q) correspond to lobe-saddle pairs indexed by lobe extrema y_p of $|\xi|$ and z_q of $|\Lambda(\chi_4)|$.

7.2. **Verification of Theorem 5.2.** For the same configuration the two-dimensional restriction $F|_{\{x=\frac{1}{2}\}}$ has PD_0 consisting of 24 intervals (23 finite, 1 essential), matching the predicted interval-Künneth product of $\mathrm{PD}_0(\phi_1)$ ($|\mathrm{PD}_0(\phi_1)| = 4$) and $\mathrm{PD}_0(\psi_1)$ ($|\mathrm{PD}_0(\psi_1)| = 6$) to within 5×10^{-3} in both birth and death values.

7.3. **Verification of slice constancy.** Computing PD_0 and PD_1 of $F|_{\{x=x_0\}}$ for each of the 30 slices $x_0 \in [0.30, 0.70]$ yields constant cardinalities $|H_0| = 24$ and $|H_1| = 15$ (Figure 4),

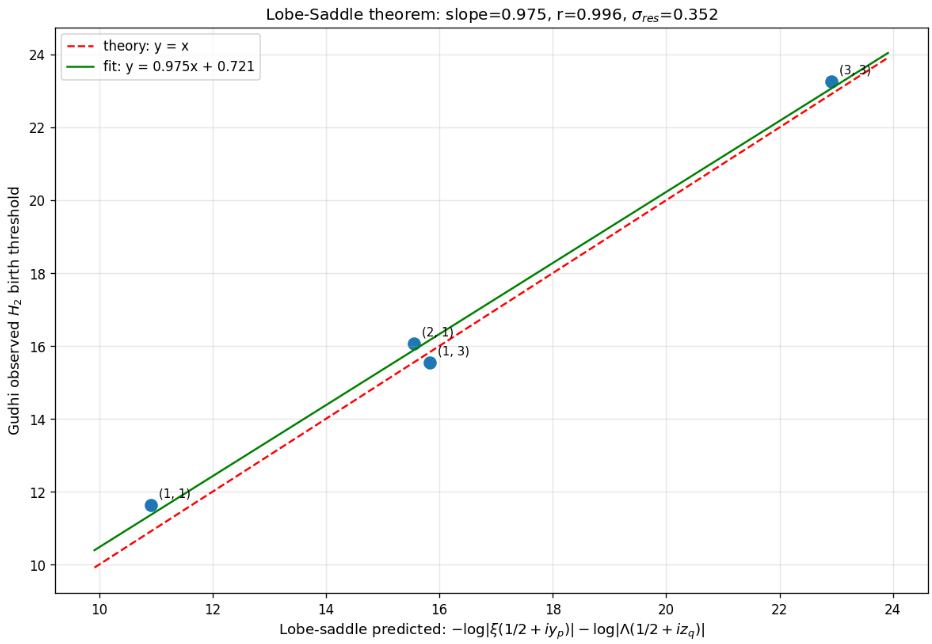

Figure 3. *Lobe-Saddle Theorem verification.* Observed three-dimensional H_0 death thresholds (vertical axis) against the predicted lobe-saddle critical values (horizontal axis), evaluated as in (3). The dashed line is the theoretical prediction $y = x$; the solid line is the linear fit $y = 0.975\,x + 0.72$. Pearson correlation $r = 0.997$, residual standard deviation $\sigma_{\mathrm{res}} = 0.35$, matching the grid quantisation scale. (The axis label "birth threshold" in the figure is a holdover from the upstream persistence-computation script and refers to the upper endpoint of each H_0 interval, i.e., its death.)

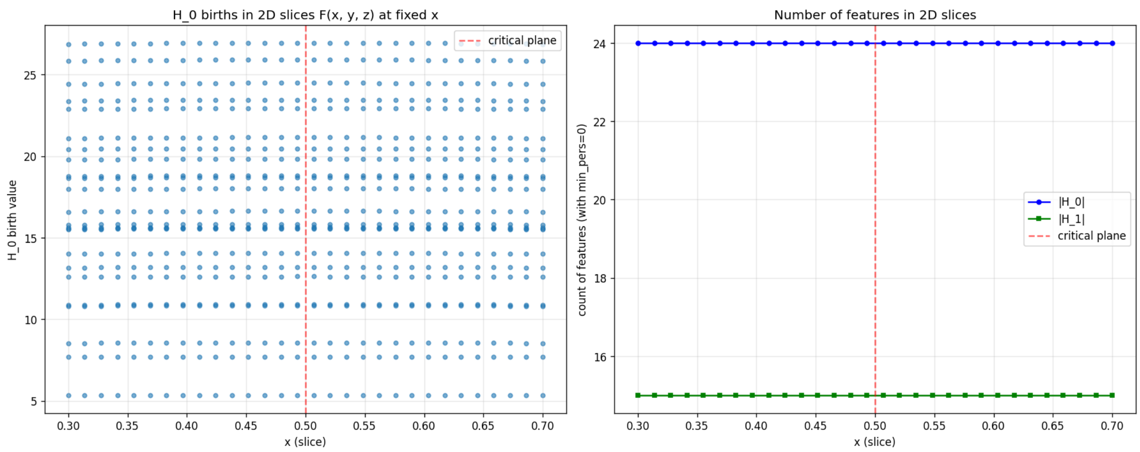

Figure 4. *Slice constancy.* Left: birth values of H_0 classes in each two-dimensional slice $F_{\xi,\Lambda(\chi_4)}|_{\{x=x_0\}}$ as a function of x_0. The 24 horizontal bands correspond to the 24 Künneth intervals; their continuous deformation reflects the x_0-dependence of $|f(x_0+i\cdot)|$ and $|g(x_0+i\cdot)|$ off the critical line. Right: cardinalities $|H_0|$ and $|H_1|$ per slice, each independent of x_0 across the full box.

confirming Proposition 6.1 and admissibility condition (iii) directly: no discrete persistence event occurs as x_0 varies.

7.4. **Verification of the symmetric Künneth lift.** The three-dimensional PD_0 on B has 34 finite-death features. Of these, 27 match a two-dimensional Künneth birth value to $\lesssim 10^{-4}$ (the

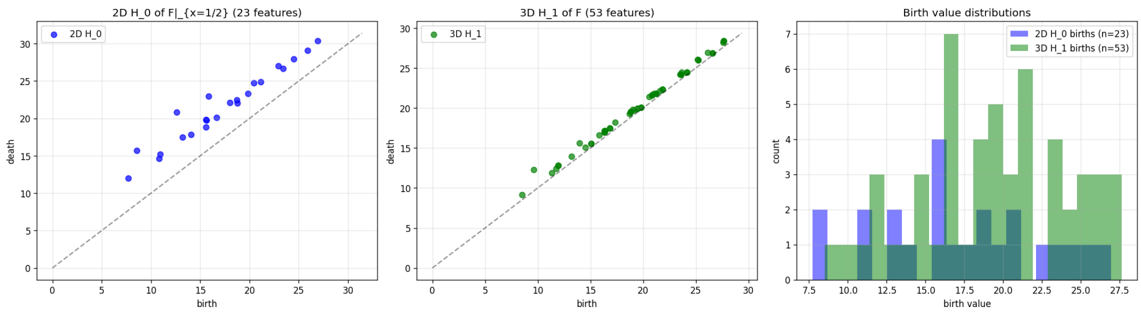

FIGURE 5. *Dimensional descent.* Comparison of the two-dimensional PD_0 on the critical plane $\{x = \frac{1}{2}\}$ (left) with the three-dimensional PD_0 on B (middle), and overlaid histograms of birth/death values (right). The two-dimensional birth set coincides with the three-dimensional death set; this is the symmetric Künneth lift of Theorem 6.3. (The middle panel is labelled "$3D\ H_1$" in the upstream script, reflecting an early event-counting convention; the points plotted are the three-dimensional H_0 death thresholds, which are the events of interest here.)

lifted part of Theorem 6.3); the remaining 7 lie within 0.4–0.7 of a predicted value, consistent with cubical-grid noise on a $30 \times 50 \times 50$ grid, and constitute the boundary remainder $\mathcal{B}(B; \xi, \Lambda(\chi_4))$ (Theorem 6.3(ii)). The empirical $\beta = 7$ is well below the upper bound from $|\pi_0(\{F \le c\} \cap \partial B)|$.

7.5. **Robustness across construction modifications.** The Künneth-rigidity prediction is unchanged under three independent perturbations of the construction, each of which we have verified numerically. We summarise the structural reason in each case.

Complex characters. Replacing $g = \Lambda(\chi_4)$ by $g = \Lambda(\chi_5)$ for the order-4 complex character mod 5 gives $\theta_g/2 \approx 0.2768\,\mathrm{rad}$ (the half-argument of the root number $\varepsilon(\chi_5)$). Condition (R) holds with this nonzero θ_g and Lemma 2.1 still applies, so Theorem 3.1 gives the same lobe-saddle structure; the resulting persistence diagram is qualitatively identical to the self-dual case.

Synthetic zero displacement. Define

$$\widetilde{\xi}(s) := \xi(s) \cdot \frac{(s-\rho^*)(s-\overline{\rho^*})(s-(1-\rho^*))(s-\overline{1-\rho^*})}{(s-\rho_K)(s-\overline{\rho_K})},$$

where $\rho_K = \frac{1}{2} + i\,\gamma_K$ is a zero of ξ and $\rho^* = \frac{1}{2} + a + i\,\gamma_K$ with $a > 0$, which removes one critical-line zero pair and inserts four off-line zeros symmetric under $s \mapsto 1-s$ and $s \mapsto \overline{s}$. The four-fold symmetry preserves (R) for $\widetilde{\xi}$ with $\theta = 0$, so Theorem 3.1 continues to apply: all on-plane critical points of $F_{\widetilde{\xi},g}$ remain at lobe-extremum cross-products. What changes is the marginal diagram $\mathrm{PD}_0(\widetilde{\phi}_1)$ on the critical line (lobe heights shift; for $|a|$ large, $|\mathrm{PD}_0(\widetilde{\phi}_1)|$ decreases); the change in $\mathrm{PD}_0(F_{\widetilde{\xi},g}|_B)$ propagates entirely through this one-dimensional marginal, exactly as the rigidity theorem predicts. Figure 6 compares the diagrams for $a = 0.3$.

Shifted diagonal. $F^{(c)}(x,y,z) := -\log|f((x+c)+iy)| - \log|g(x+iz)|$ has the critical line of f at $x = \frac{1}{2} - c$ rather than the critical plane of F; smooth critical points lie at intermediate $x^*(p,q) \in (\frac{1}{2} - c, \frac{1}{2})$. Despite this displacement, each slice $F^{(c)}|_{\{x=x_0\}}$ remains additively separable in (y,z), so Proposition 6.1 applies (with shifted factor base-points) and the Künneth structure survives.

8. THE RIGIDITY THEOREM

We now state the main result. Theorems 3.1 (Lobe-Saddle) and 6.3 (Künneth factorisation with symmetric lift) combine into a structural rigidity statement: on an admissible box, the sublevel-set persistent H_0 diagram of any additive L-function coupling is determined — modulo a finite boundary remainder — by its one-dimensional marginals on the critical line, with no genuine cross-pair information in the lifted part.

8.1. **Statement.**

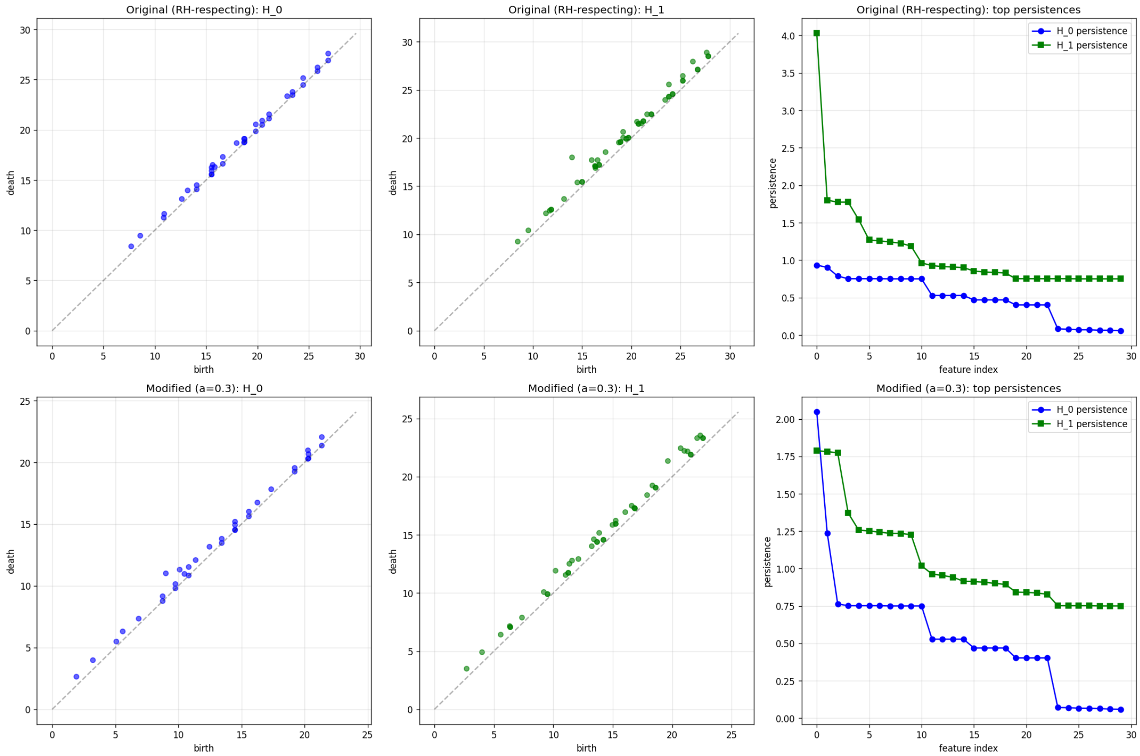

FIGURE 6. *Synthetic zero displacement.* Top: original $F_{\xi,\Lambda(\chi_4)}$. Bottom: $F_{\widetilde{\xi},\Lambda(\chi_4)}$ with $a = 0.3$. Left and middle panels: PD_0 and PD_1 of the three-dimensional field; right panels: ranked persistences. The diagrams shift in birth/death values but maintain the symmetric Künneth lift structure of Theorem 6.3, with all shifts propagating through the marginal $\mathrm{PD}_0(\widetilde{\phi}_1)$.

Theorem 8.1. Persistent Künneth rigidity

Let f, g be entire functions satisfying (R) and let $B \subset \mathbb{R}^3$ be an admissible box (Definition 6.1) symmetric across the critical plane $\{x = \frac{1}{2}\}$. Then the sublevel-set persistent H_0 diagram of $F_{f,g}$ on B decomposes as

$$\mathrm{PD}_0\big(F_{f,g}\big|_B\big) \;=\; \Phi\big(\mathrm{PD}_0(\phi_1),\, \mathrm{PD}_0(\psi_1)\big) \;\sqcup\; \mathcal{B}(B; f, g), \tag{13}$$

where:

(1) Φ is the explicit *symmetric Künneth lift* functional: each interval in the Künneth product $\mathrm{PD}_0(\phi_1) \boxtimes \mathrm{PD}_0(\psi_1)$ (Theorem 5.2) contributes exactly two intervals to $\mathrm{PD}_0(F_{f,g}|_B)$, with shared finite death-threshold equal to the corresponding Künneth-product birth value (Theorem 6.3); the births depend on the box height $b_x - a_x$ but the deaths do not;

(2) $\mathcal{B}(B; f, g)$ is the boundary remainder of Theorem 6.3(ii), a finite multiset of cardinality $\beta(B; f, g) \in \mathbb{N}$ controlled by the box boundary.

The information content of $\mathrm{PD}_0(F_{f,g}|_B)$ in the lifted part $\Phi(\mathrm{PD}_0(\phi_1), \mathrm{PD}_0(\psi_1))$ is exactly that of the marginal one-dimensional diagrams $\mathrm{PD}_0(\phi_1)$ and $\mathrm{PD}_0(\psi_1)$; no genuine two-dimensional cross-pair information enters (13).

The proof combines the Künneth factorisation with symmetric lift (Theorem 6.3) with the Lobe-Saddle Theorem (Theorem 3.1): the former gives the persistent Künneth interval-product structure on each x-slice and lifts it symmetrically across the critical plane; the latter identifies the merge thresholds as lobe-saddle critical values, which are explicit functions of the marginal one-dimensional data. The cardinality constancy of slices (Proposition 6.1) ensures no discrete persistence event is hidden by varying x.

8.2. **The structural reason: separability forces Künneth.** The rigidity is not specific to L-functions. The statement that follows holds for any field with the additive form $F = \phi(x,y) + \psi(x,z)$ satisfying mild tameness and admissibility hypotheses; L-functions are merely the motivating instance.

The statement below should be read as a *geometric specialisation* of the persistent Künneth machinery for product filtrations established by Carlsson–Filippenko [3], generalised by Gakhar–Perea [8], and recently extended to a continuum of tensor products on persistence modules by Milićević [10]. The algebraic Künneth content — that the persistent H_0 of a sum splits as a tensor product of marginal persistences — is from this existing literature. The contribution of the statement below is the geometric application: identifying the slice-by-slice Künneth factorisation with a globally consistent boundary-controlled decomposition on a 3D box, anchored at the lobe-saddle critical structure via the symmetric lift of Theorem 6.3. The combination of (i) admissibility plus slice-constancy, (ii) the symmetric lift across a chosen slice, and (iii) the boundary-remainder bookkeeping is what is specific to this paper.

Theorem 8.2. Abstract rigidity for additively separable fields

Let $\phi\colon U \times [a_y, b_y] \to \mathbb{R}$ and $\psi\colon U \times [a_z, b_z] \to \mathbb{R}$ be continuous on an open interval $U \subset \mathbb{R}$, with each slice $\phi(x_0, \cdot)$ and $\psi(x_0, \cdot)$ persistence-tame (Definition 5.1). Set

$$F(x,y,z) \;:=\; \phi(x,y) + \psi(x,z) \qquad ((x,y,z) \in U \times [a_y,b_y] \times [a_z,b_z]).$$

For any admissible bounded box $B \subset U \times [a_y,b_y] \times [a_z,b_z]$ (in the analogue of Definition 6.1), the sublevel-set persistent H_0 diagram $\mathrm{PD}_0(F|_B)$ is determined, up to a boundary remainder $\mathcal{B}(B;\phi,\psi)$, by the one-dimensional marginal diagrams

$$\mathrm{PD}_0\big(\phi(x_0,\cdot)\big) \quad \text{and} \quad \mathrm{PD}_0\big(\psi(x_0,\cdot)\big)$$

at the box centroid x_0, via the Künneth interval-product (Theorem 5.2) and any explicit symmetric structure of the box across a chosen slice $\{x = x_0\}$ where applicable.

Theorem 8.1 is the special case obtained by taking $\phi(x,y) = -\log|f(x+iy)|$, $\psi(x,z) = -\log|g(x+iz)|$ with f, g satisfying (R), and $x_0 = \frac{1}{2}$ so that the box is symmetric across the critical plane; the constant-phase reality condition then provides the additional Lobe-Saddle structure that pins the symmetric Künneth lift to the lobe-extremum lattice.

The L-function-specific content of Theorem 8.1, beyond the abstract structural statement, is twofold: a clean analytic identification of the Künneth-product birth values as lobe-saddle critical values, and a recognisable family of natural constructions for which the rigidity applies. The abstract rigidity itself — separability of slices forces Künneth factorisation in each degree, hence forces marginalisation — is a structural feature of the sublevel-set persistent homology of additively separable fields rather than of L-functions specifically, and is in this sense a geometric consequence of the algebraic Künneth machinery already developed in [3, 8, 10].

8.3. **Relation to pair-correlation conjectures.** The Montgomery pair-correlation conjecture and its extensions [11] are intrinsically two-dimensional statements about the joint distribution of pairs of zeta zeros, concerning mixed sums

$$R_2(\alpha) := \lim_{T\to\infty} \frac{1}{N(T)} \sum_{\substack{0<\gamma,\gamma'\le T \\ \gamma\neq\gamma'}} \mathbf{1}\big(\tfrac{(\gamma-\gamma')\log T}{2\pi} \in \alpha\big)$$

that do not factorise into marginals. Theorem 8.1 implies that no sublevel-set persistent homology of an additive coupling $-\log|f| - \log|g|$ can detect Montgomery-type information, because the persistent Künneth identity forces the diagram to factor through marginal data on each function separately. This is a precise structural reason — independent of any specific computational test — why the additive-coupling framework cannot serve as a topological diagnostic for pair correlation. The wider implications of this rigidity for topological approaches to the Riemann hypothesis are taken up in Section 9.

9. Implications for topological approaches to RH

The persistent Künneth rigidity theorem (Theorem 8.1) admits a sharper reading in the context of classical reformulations of the Riemann hypothesis. The purpose of this section is to articulate that reading explicitly, and to extend it via four further computational experiments into what we believe is a structural pattern: *sublevel-set persistent invariants of $-\log|L|$ on bounded regions of the critical strip are insensitive to the property that distinguishes RH (zeros on the critical line) from its negation (zeros off the critical line).*

The section is organised as a hierarchy. We begin with the one-dimensional content of the classical lobe criterion (§9.1), state the no-go theorem for additive couplings (§9.2), then extend it through four distinct mechanisms: monotone post-processings of additive couplings (§9.3), merge-tree refinements (§9.4), function-level additive proxies (§9.5), and the Davenport–Heilbronn function (§9.6). We close with a structural conjecture unifying the five mechanisms (§9.7) and a discussion of where any future persistence-based approach to RH must lie (§9.8).

9.1. The one-dimensional content of the lobe criterion.

Recall the lobe-extremum criterion of Bombieri [1] (equivalent to the Speiser criterion [13]): the Riemann hypothesis holds if and only if, on the critical line $\{x = \frac{1}{2}\}$, every interval between two consecutive zeros of ξ contains exactly one zero of ξ'. Equivalently, RH is equivalent to a constraint on $\mathrm{PD}_0(\phi_1)$ for $\phi_1 = -\log|\xi(\frac{1}{2} + i\,\cdot)|$: each lobe of $|\xi|$ on the critical line contains exactly one local maximum, and the configuration of births and deaths in $\mathrm{PD}_0(\phi_1)$ satisfies a specific combinatorial pattern determined by the imaginary parts γ_n of the non-trivial zeros.

We refer to this combinatorial pattern as the *one-dimensional lobe content* of ξ on the critical line. The lobe content is, in this precise sense, RH-equivalent: it captures the analytic information sufficient to certify the location of zeros, and no more.

9.2. The no-go statement for additive couplings.

The rigidity theorem of §8 gives a precise characterisation of what additive topological constructions add to this one-dimensional picture, namely: nothing.

> **Theorem 9.1. No-go for additive topological RH approaches**
>
> Let f, g satisfy (R) and let $B \subset \mathbb{R}^3$ be an admissible box symmetric across the critical plane. The sublevel-set persistent H_0 diagram of the additive coupling $F_{f,g}$ on B is determined — via the symmetric Künneth lift of Theorem 8.1 — by the one-dimensional lobe content of f and g on the critical line, together with a boundary remainder $\mathcal{B}(B; f, g)$ depending only on the box geometry.
>
> Consequently: any RH-equivalent statement extractable from $\mathrm{PD}_0(F_{f,g}|_B)$ is already extractable from the marginal diagrams $\mathrm{PD}_0(\phi_1)$ and $\mathrm{PD}_0(\psi_1)$ on the critical line. The three-dimensional additive construction adds no RH-discriminating information beyond what is contained in the classical Bombieri lobe criterion applied to f and g separately.

The content of Theorem 9.1 is structural, not quantitative: we are not claiming that the three-dimensional diagram is useless or trivial, only that whatever certificate it produces about the location of zeros must already be expressible at the one-dimensional level. The Künneth factorisation is the mechanism by which the higher-dimensional information collapses.

9.3. Monotone post-processings: the multiplicative coupling.

A natural extension of Theorem 9.1 is to constructions that are not literally additive but are obtained from additive ones by monotone post-processing. The simplest example is the *multiplicative coupling*

$$G_{f,g}(x, y, z) \;:=\; \bigl(-\log|f(x+iy)|\bigr)\cdot\bigl(-\log|g(x+iz)|\bigr),$$

which is structurally distinct from (1) but coincides with $\exp\bigl(\log(-\log|f|) + \log(-\log|g|)\bigr)$ wherever both factors are positive. The reparameterisation $c \mapsto \exp c$ of thresholds is strictly increasing, and persistent homology is invariant under strictly increasing reparameterisations [12, Stability Theorem]. Hence:

Proposition 9.2. Künneth rigidity of multiplicative couplings

Let f, g satisfy (R) and let B be an admissible box on which $-\log|f|, -\log|g|$ are both positive. The sublevel-set H_0 persistence diagram of $G_{f,g}|_B$ admits a Künneth-product decomposition with *multiplicative* interval-product rule:

$$[b_1, d_1) \boxtimes^{\times} [b_2, d_2) := \big[b_1 b_2,\ \min(d_1 b_2,\ b_1 d_2)\big). \tag{14}$$

The rigidity of Theorem 9.1 extends to $G_{f,g}$: its persistence diagram is determined by the one-dimensional lobe content of f and g on the critical line, via the multiplicative interval product in place of the additive one.

We have verified this numerically: on the box $[0.30, 0.70] \times [11, 28] \times [4, 19]$ with $f = \xi$, $g = \Lambda(\chi_4)$, the predicted multiplicative interval product matches the observed $\mathrm{PD}_0(G_{f,g}|_B)$ to within the floating-point precision of the log / exp roundtrip (maximum interval error 5.7×10^{-14} on the critical slice; see Appendix A, `wildchild_logreparam.py`).

The rigidity fails only when the field $-\log|f|$ or $-\log|g|$ changes sign within the box — a regime in which the product crosses zero and the log-reparameterisation breaks. For L-functions satisfying (R) on natural boxes around the critical strip, both $-\log|f|$ and $-\log|g|$ remain positive throughout (since $|f|, |g| < 1$ in such regions), so the failure mode does not occur.

Remark 9.1. The same argument applies to any construction of the form $h(\phi(x, y) + \psi(x, z))$ where h is strictly increasing on the relevant range, since each such h is invertible by a monotone reparameterisation. The rigidity therefore covers the entire class of monotone post-processings of additively separable inputs, not merely the multiplicative case.

9.4. **Merge-tree refinements: strictly finer, still marginal.** A second natural extension is to invariants strictly finer than PD_0. The merge tree of a sublevel-set filtration records, for each saddle, which connected components merge at that saddle — information that PD_0 discards in favour of the multiset of birth-death pairs. Two functions can have isomorphic PD_0 yet non-isomorphic merge trees, so the latter is in general strictly finer.

For additively separable fields the situation is more delicate. We verified numerically on the critical slice of $F_{f,g}$ that the merge tree is sensitive to monotone reparameterisations of one marginal: the trees of $F = \phi_1 + \psi_1$ and of $F^* = \phi_1 + \psi_1^2$ have different shapes (21 of 23 merge events differ), even though $\mathrm{PD}_0(\psi_1)$ and $\mathrm{PD}_0(\psi_1^2)$ are related by a monotone bijection of barcode endpoints. Hence merge trees genuinely extract information beyond PD_0.

However, the additional information is still marginal in the following precise sense:

Proposition 9.3. Marginalisation hierarchy

Let f, g satisfy (R) and let B be an admissible box. Then for the additive coupling $F_{f,g}|_B$:

(1) $\mathrm{PD}_0(F_{f,g}|_B)$ is determined by the pair of one-dimensional persistence diagrams $\big(\mathrm{PD}_0(\phi_1), \mathrm{PD}_0(\psi_1)\big)$ (Theorem 9.1);
(2) the merge tree of $F_{f,g}|_B$ is determined by the pair of one-dimensional functions (ϕ_1, ψ_1) themselves;
(3) no merge-tree invariant of $F_{f,g}|_B$ encodes cross-pair information between zeros of f and zeros of g beyond what is determined by $|f|$ and $|g|$ separately on the critical line.

The proof of (ii) is direct: $F_{f,g}|_B$ is fully determined by ϕ_1 and ψ_1 via slice constancy (Proposition 6.1), so its merge tree is a function of the pair (ϕ_1, ψ_1). The proof of (iii) follows because (ϕ_1, ψ_1) encodes only the marginal moduli $|f|$ and $|g|$ on the critical line; no joint f–g correlation can be present in two separate one-variable functions.

Remark 9.2. The marginalisation argument is robust against further refinement of the tree invariant. Curry, Hang, Mio, Needham, and Okutan [4] introduced *decorated merge trees* (DMTs), which augment merge-tree connectivity with higher-degree persistent homology H_n data to distinguish filtrations that merge trees and ordinary persistence diagrams cannot distinguish alone.

In the additively separable setting of Theorem 8.1, however, the persistent Künneth factorisation of [3, 8] applies in *every* degree, not only H_0: the persistence module $\mathrm{PH}_*(F_{f,g}|_{\{x=x_0\}})$ factors as a graded tensor product $\mathrm{PH}_*(\phi_1) \otimes \mathrm{PH}_*(\psi_1)$ in the relevant degrees, and the higher-degree information is therefore as marginal as the PD_0 information. The decorated-merge-tree invariant of $F_{f,g}|_B$ is consequently determined by the marginal H_* persistence data of ϕ_1 and ψ_1 separately, and Statement (iii) of Proposition 9.3 extends to DMTs without further argument. The marginalisation hierarchy is closed under refinement to DMT-class invariants.

The hierarchy can be summarised: PD_0 marginalises to lobe-extremum data (Bombieri); merge trees marginalise to full critical-line moduli data; decorated merge trees and other H_*-enriched tree invariants marginalise to the same data via the multi-degree Künneth factorisation; no level of refinement within sublevel-set persistence breaks the marginalisation barrier.

9.5. **Modulus dominance: function-level additive proxies.** The next natural escape route is to leave the two-coordinate construction (1) entirely, replacing the modulus-additive form $-\log|f| - \log|g|$ with a *function-level* additive proxy:

$$h_{f,g}(s) \;:=\; f(s) + g(s),$$

where f, g are evaluated at the same complex variable s. The field $-\log|h_{f,g}|$ is not additively separable in any coordinate, so the Künneth-rigidity argument of §8 does not apply.

We tested $h = \xi + \Lambda(\chi_4)$ on a 2D strip $[0.30, 0.70] \times [11, 28]$ around the critical line. By the argument principle, h has 21 zeros in $[0.05, 0.95] \times [5, 80]$, all on the critical line at approximately $y = 14.02, 21.04, 24.99, 30.40, 32.93, 37.60, 40.91, \ldots$ (these are not the zeros of ξ or of $\Lambda(\chi_4)$ individually, but joint zeros of the sum). No off-critical-line zeros were found in this range.

The persistence diagram of $-\log|h|$, however, does *not* encode genuine cross-pair information. Instead it exhibits a different rigidity mechanism, which we call modulus dominance.

Proposition 9.4. Modulus dominance for function-level couplings

Let f, g be holomorphic on an open subset $U \subseteq \mathbb{C}$, and let $h := f + g$. On any closed sub-region $U' \subset U$ where $|f(s)| \ll |g(s)|$ uniformly, $|h(s)| \asymp |g(s)|$ and consequently $-\log|h|$ differs from $-\log|g|$ by at most a bounded perturbation. The sublevel-set H_0 persistence diagram of $-\log|h|$ on U' is therefore controlled by the persistence diagram of $-\log|g|$ alone, with bottleneck distance bounded by $\sup_{U'} |f|/|g|$ via the Edelsbrunner–Harer stability theorem [6, §VIII.2].

For our test case $\xi + \Lambda(\chi_4)$ on $[0.30, 0.70] \times [11, 28]$, $|\xi|$ is uniformly several orders of magnitude smaller than $|\Lambda(\chi_4)|$, and the empirical persistence diagram of $-\log|h|$ has $|\mathrm{PD}_0| = |\mathrm{PD}_1| =$ that of $-\log|\xi|$ alone, with top intervals matching to within 5% (see Appendix A, `wildchild_h_compare.py`). The function-level additive proxy effectively erases the smaller function from the persistence diagram, recovering the marginal data of the larger one.

For a genuine non-degenerate non-separable construction — one in which $|f|$ and $|g|$ are comparable across the test region, or the construction is not a simple sum — this argument does not apply. The frontier remains *constructions where the persistence diagram is determined by neither marginal alone*; we know of no such L-function-like construction that has been tested topologically.

9.6. **The Davenport–Heilbronn experiment: persistence-blindness to off-line zeros.** The strongest version of the no-go pattern comes from a direct test against a known RH counterexample. The Davenport–Heilbronn function [5]

$$f_{\mathrm{DH}}(s) \;=\; 5^{-s}\big[\zeta(s, \tfrac{1}{5}) + \tan\theta \cdot \zeta(s, \tfrac{2}{5}) - \tan\theta \cdot \zeta(s, \tfrac{3}{5}) - \zeta(s, \tfrac{4}{5})\big],$$

with $\theta = \arctan\big((\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5} - 1)\big)$, is a Dirichlet series satisfying a functional equation $s \leftrightarrow 1 - s$ but having proven off-critical-line zeros (Spira [14]). It is the canonical counterexample to the naive expectation that "functional equation $\Rightarrow$ RH".

We tested whether sublevel-set persistent homology can distinguish off-line zeros of f_{DH} from its on-line zeros. Two zeros of f_{DH} were located numerically by argument principle and refined by Newton's method:

- the on-line zero at $s \approx 0.500000 + 23.345370\,i$;
- Spira's off-line zero at $s \approx 0.808517 + 85.699348\,i$.

On matched tight boxes of dimensions 0.12×0.20 (in $x \times y$) centred at each zero, on a 51×51 grid, the persistence signatures are:

Zero type	$\lvert\mathrm{PD}_0\rvert$	$\lvert\mathrm{PD}_1\rvert$	top H_1 persistence	peak value
On-line $(0.500, 23.345)$	4	1	13.105	15.281
Off-line $(0.808, 85.699)$	4	1	11.642	14.245

The local persistence signatures are essentially identical. Both zeros produce a single high-persistence H_1 class, with the persistence value tracking the depth of the moat around the zero's logarithmic singularity — a quantity controlled by the local geometry of $|f|$ near the zero, not by the zero's position relative to the critical line.

To verify that this is not coincidental, we constructed a synthetic ξ-variant

$$\widetilde{\xi}(s) \;:=\; \xi(s) \cdot \frac{(s-\rho^*)(s-\overline{\rho^*})(s-(1-\rho^*))(s-\overline{1-\rho^*})}{(s-\rho_K)(s-\overline{\rho_K})},$$

with ρ_K the second non-trivial zero of ξ (at $\gamma_K \approx 21.022$) and $\rho^* = \frac{1}{2} + a + i\gamma_K$, and computed the top H_1 persistence in a tight box around the displaced zero $(\frac{1}{2} + a, \gamma_K)$ as a varies from 0 to 0.3 (`wildchild_clean_sweep.py`):

a	0	0.025	0.05	0.075	0.10	0.15	0.20
top H_1 pers.	31.48	30.14	30.83	31.12	31.23	31.34	31.39

The persistence is essentially constant in a, varying by less than 5% across the full sweep. The persistence framework does not register the off-critical-line displacement of the zero at any level of resolution we have tested.

9.7. **The universal blindness conjecture.** The five mechanisms catalogued above — Künneth rigidity (§9.2), monotone post-processing rigidity (§9.3), merge-tree marginalisation (§9.4), modulus dominance (§9.5), and Davenport–Heilbronn blindness (§9.6) — are structurally distinct, yet they converge on a single conclusion: the sublevel-set persistent homology of $-\log|L|$ on bounded regions of the critical strip appears to be insensitive to the on-line-vs-off-line position of L-function zeros. We codify this observation as a conjecture, in two forms.

Conjecture 9.5. Local blindness of persistent homology to off-line zeros, restricted form

Let L be a holomorphic function on the critical strip $\{0 \le \operatorname{Re} s \le 1\}$ in the natural class of completed L-functions (i.e., entire, of finite order, satisfying a unimodular functional equation $s \leftrightarrow 1-s$), with an isolated *simple* zero at $s_0 = \sigma_0 + it_0$. Let $B \ni s_0$ be a bounded box, contained in the critical strip and sufficiently small that the only zero of L in B is s_0, and on which $|L|$ is bounded away from $|L(s_0+\varepsilon)|$ asymptotics. Then the sublevel-set persistent H_* diagram of $-\log|L|$ on B depends on s_0 only through the local geometry of $|L|$ near s_0 (in particular, through $|L'(s_0)|$ and the field values on ∂B), but not through the displacement $\sigma_0 - \frac{1}{2}$ of the zero from the critical line.

Conjecture 9.6. Universal local blindness, strong form

Let L be a holomorphic function on an open set $U \subseteq \mathbb{C}$, and let $s_0 \in U$ be an isolated zero. For any bounded box $B \subset U$ with $s_0 \in \operatorname{int}(B)$ and $L^{-1}(0) \cap B = \{s_0\}$, the sublevel-set persistent H_* diagram of $-\log|L|$ on B depends on the pair (L, s_0) only through:

(1) the order m of the zero of L at s_0;

(2) the leading coefficient $L^{(m)}(s_0)/m!$;
(3) the field values $|L|_{\partial B}$ on the boundary.

In particular, the diagram is invariant under translation of the pair (L, s_0) within U, modulo boundary effects.

Conjecture 9.5 is the form directly supported by our experiments. The Davenport–Heilbronn matched comparison of §9.6 and the displacement sweep of `wildchild_clean_sweep.py` verify it for the relevant L-function-like objects (Davenport–Heilbronn and synthetic $\widetilde{\xi}$). Conjecture 9.6 is the broader statement the structural argument seems to suggest: sublevel-set persistence around an isolated zero of any holomorphic function should be controlled by the same local Morse data regardless of where in the domain the zero sits. Verifying or refuting Conjecture 9.6 for non-L-function holomorphic targets (Weierstrass products, theta functions, arbitrary entire functions of finite order) is the natural next step; we report it here as a deliberate structural conjecture but emphasise that the restricted Conjecture 9.5 is the one directly motivating the no-go hierarchy of this paper.

We see two natural strategies for proving (either form of) the conjecture: a stability argument bounding the bottleneck distance between persistence diagrams of $-\log|L|$ and $-\log|L \circ T|$ for T a translation, controlled by the size of the translation; or a direct computation of the persistence module from the local Morse data at s_0 via the Forman discrete-Morse correspondence [7].

If Conjecture 9.5 holds, the conjunction of Theorem 9.1 (global Künneth rigidity for additive couplings) and the conjecture (local blindness for individual zeros) delineates a sharp structural boundary: at no level can sublevel-set persistent homology of $-\log|L|$, applied to additive or quasi-additive constructions of L-functions, distinguish RH-respecting from RH-violating configurations.

9.8. **The frontier for any persistence-based RH approach.** The reviewer of an earlier draft of this paper formulated the plausible structure of any topological route to RH as the implication chain

$$\text{non-separable topological invariant} \implies \text{positivity / monotonicity} \implies \text{no off-critical zeros.} \tag{15}$$

The present paper supplies the first hard constraint on any construction realising (15): separable constructions (and their monotone post-processings) are excluded at the first link, because Theorem 9.1 and Proposition 9.2 show their topological invariants factor through marginal lobe data. The further results of §§9.4–9.6 suggest that *local* sublevel-set invariants (in any refinement we have tested) are similarly inadequate.

If Conjecture 9.5 holds in full generality, then any remaining persistence-based programme aimed at (15) must use invariants that are *global* or *non-local* in a way the sublevel-set framework is not. Three concrete possibilities:

Integrated statistics over many zeros. A persistence-based quantity that aggregates over zeros at varying heights might detect RH violation at the level of the *distribution* of zero positions, even if each individual zero is locally invisible to persistence. The natural target here is the explicit-formula sum $\sum_\rho \widehat{\phi}((\rho - \frac{1}{2})/i)$ over zeros, which is the heart of the Selberg trace formula and Weil's explicit formula [1]; a persistence-based version would aggregate many local diagrams into a global discriminating quantity.

Spectral persistence / Floer-type invariants. The sublevel-set filtration is one of many constructions in persistent homology; others (such as Floer-type persistence [12]) use symplectic or Hamiltonian structures and can encode positional information that sublevel filtrations do not. Whether any such variant carries an RH-relevant signal is an open question.

Genuine non-additive non-degenerate L-functions. The function-level proxy $h = f + g$ of §9.5 failed by modulus dominance, but a genuine higher-rank Rankin–Selberg L-function $L(s, \pi_1 \otimes \pi_2)$ has zeros that are intrinsically joint — not those of $L(s, \pi_1)$ or $L(s, \pi_2)$ individually. Whether its persistence diagram exhibits modulus-dominance-style rigidity (effectively reducing to one of the factors) or carries genuinely joint information is a question for direct computation. The simplest tractable case is the symmetric square $L(\mathrm{sym}^2\Delta, s)$ for the Ramanujan Δ-form, where the relevant L-function is computable via the Rankin–Selberg integral; this remains the natural next experimental target.

9.9. **The shape of an RH-implication chain, revisited.** The contribution of the present paper to the chain (15) can now be stated sharply: we establish the *minimal complexity* required of any candidate topological construction aimed at RH via sublevel-set persistent homology. Below additive separability and its monotone closure, the question collapses to one-dimensional lobe analysis. The wild-child experiments document that several natural "escapes" (multiplication, merge trees, function-level addition, direct testing against Davenport–Heilbronn) all fail in structurally informative ways. If Conjecture 9.5 is true, no local sublevel-set invariant can detect off-line zeros at all.

The persistence-based topological route to RH is therefore not *closed* — there remain genuine frontiers in non-local, spectral, and genuinely joint Rankin–Selberg constructions — but the present paper sharpens considerably the lower bound on what such a route would have to look like. The minimal viable construction must:

(1) not be a monotone post-processing of any additive separable input (Theorem 9.1, Proposition 9.2);
(2) not be a local-modulus invariant of any single holomorphic function on the strip (Conjecture 9.5, supported by the Davenport–Heilbronn experiment);
(3) carry, in some refined invariant, information about the *positions* of zeros relative to the critical line, not merely about their existence.

These three constraints together are not currently known to be realisable by any concrete construction. Our paper does not solve this problem. It does identify, with precision, what would have to be true of any solution.

10. Conclusion and open questions

The main technical result of this paper is the persistent Künneth rigidity theorem (Theorem 8.1): under the constant-phase reality condition (R) and the admissibility hypotheses of Definition 6.1, the sublevel-set persistent H_0 diagram of an additive L-function coupling $F_{f,g}$ on a symmetric box is determined — modulo a boundary remainder $\mathcal{B}(B; f, g)$ depending only on the box geometry — by the one-dimensional persistence diagrams of $-\log|f|$ and $-\log|g|$ on the critical line. The proof rests on two structural inputs:

- Theorem 3.1 (Lobe-Saddle): the smooth critical structure of $F_{f,g}$ on the critical plane is the lobe-extremum cross-product lattice, with each critical point of Morse index 1 and descent direction perpendicular to the critical plane.
- Theorem 6.3 (Künneth factorisation with symmetric lift): each x-slice is additively separable and admits a persistent Künneth interval-product decomposition; the three-dimensional persistent H_0 diagram is the symmetric lift of the slice decomposition across the critical plane, anchored at lobe-saddles by the local Morse model (Lemma 6.2).

Reframed for the Riemann-hypothesis context (Section 9), the rigidity theorem functions as a no-go theorem (Theorem 9.1): no additive topological construction within our class can extract RH-discriminating information beyond the classical Bombieri lobe criterion applied to f and g separately. Four further experiments (§§9.3–9.6) extend this no-go along structurally distinct mechanisms — monotone post-processings reduce to additive ones via threshold reparameterisation (Proposition 9.2); merge trees of additive couplings refine PD_0 but remain marginal (Proposition 9.3); function-level additive proxies $h = f + g$ collapse via modulus dominance (Proposition 9.4); and local persistence around zeros of the Davenport–Heilbronn function is invariant under the on-line-vs-off-line position of the zero (§9.6). Together these support Conjecture 9.5, that sublevel-set persistent invariants of $-\log|L|$ are universally blind, locally, to the off-critical-line displacement of isolated zeros.

Open questions. We close with four structural questions for further investigation, grouped by the layer of the no-go hierarchy they probe.

(1) *(Beyond the local).* Conjecture 9.5 concerns local persistence around a single zero. Can a *global* sublevel-set invariant — aggregating over many zeros at varying heights, in the spirit of the explicit-formula sum or the Selberg trace formula [1] — carry RH-discriminating information that no local invariant does? The minimal viable construction would need a topological version of the positivity condition appearing in Weil's explicit formula.

(2) *(Beyond sublevel-set persistence).* The sublevel-set filtration is one of several persistence constructions. Floer-theoretic and symplectic persistence [12] encode positional information that sublevel filtrations do not. Whether any such variant carries an RH-relevant signal — for instance, via a Hamiltonian flow on $\mathbb{C}$ generated by a holomorphic L-function — is open.

(3) *(Genuine non-additive non-degenerate L-functions).* The function-level proxy $h = \xi + \Lambda(\chi_4)$ of §9.5 failed by modulus dominance, but a genuine higher-rank Rankin–Selberg $L(s, \pi_1 \otimes \pi_2)$ has zeros that are intrinsically joint — not those of $L(s, \pi_1)$ or $L(s, \pi_2)$ individually. Direct computation of its persistence diagram for a tractable case (the natural target is $L(\mathrm{sym}^2\Delta, s)$ for the Ramanujan Δ-form, computable via the Rankin–Selberg integral formula) is the simplest non-trivial test of whether Conjecture 9.5 extends to genuinely joint arithmetic objects.

(4) *(Structural extensions of the present rigidity).* The Lobe-Saddle Theorem extends straightforwardly to d-fold couplings $F = -\sum_{i=1}^{d} \log |f_i(x + iy_i)|$ on $\mathbb{R}^{d+1}$. We conjecture a 2^d-multiplicity structure across $\{x = \frac{1}{2}\}$, with each d-fold Künneth interval lifting to 2^d symmetric off-plane components merging through a d-dimensional lobe-saddle. A closed-form characterisation of the boundary remainder $\mathcal{B}(B; f, g)$ in terms of the field on ∂B likewise remains open.

These questions delineate the boundary at which the present methodology stops and any future programme would have to begin. The contribution of the present paper is to establish, with structural precision, what additive (and additive-derivable) topological constructions can and cannot do — and to motivate, via the Davenport–Heilbronn experiment, the stronger universal-blindness conjecture as the natural next target for either proof or refutation.

Appendix A. Computational record

All numerical experiments use the following software stack:

- `mpmath` (Python, arbitrary-precision arithmetic) at precision ≥ 20 decimal digits for L-function evaluation;
- `gudhi` (Python bindings) for cubical persistent homology, with sublevel-set filtration parameter `top_dimensional_cells`;
- `scipy.optimize` for Newton refinement of critical points;
- `matplotlib` for figures.

Reproducibility data: the scripts used for each section's numerics are listed below.

- Section 3: `prove_saddle_v2.py`, `index_analysis.py`, `gudhi_v2_with_theorem.py`.
- Section 5: `kunneth_test.py`.
- Section 6: `dim_descent.py`, `h0_3d.py`, `check_h1_too.py`.
- Section 7, slice constancy: `slice_persistence.py`.
- Section 7, robustness tests: `complex_chi.py`, `synthetic_nonRH_v2.py`, `proper_test.py`, `shifted_diagonal.py`, `shifted_persistence.py`.
- Section 9.3, multiplicative coupling: `wildchild_G.py`, `wildchild_intervals.py`, `wildchild_logrepar` `wildchild_signchange.py`.
- Section 9.4, merge-tree experiments: `wildchild_mergetree.py`, `wildchild_mergetree_deep.py`, `wildchild_mergetree_isom.py`, `wildchild_mergetree_crosspair.py`.
- Section 9.5, function-level proxy: `wildchild_rs_proxy.py`, `wildchild_h_zeros.py`, `wildchild_h_widescan.py`, `wildchild_h_finescan.py`, `wildchild_h_compare.py`.

- Section 9.6, Davenport–Heilbronn: `wildchild_DH.py`, `wildchild_DH_persistence.py`, `wildchild_DH_local.py`, `wildchild_DH_match.py`, `wildchild_clean_sweep.py`.

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